



INDIAN ECONOMIC DEVELOPMENT

Higher/Senior Secondary Classes (All Streams) | 2024 |

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Dear Teachers and Students,

The study of economics focusses on how societies distribute limited resources to satisfy their limitless desires. Understanding economic development is essential for responsible citizenship and sound decision-making in the Indian context. The goal of this note pack is to give teachers and students a thorough understanding of India's economic development.

Key Features of the Notes:

- *Comprehensive coverage and exam-oriented: : The notes cover all essential topics. It is designed in alignment with the NCERT syllabus; these notes are an invaluable tool for exam preparation, including competitive examinations. They include key points, summaries, and practice questions to aid in effective revision. These notes are organised in a logical sequence, making them easy to enhance comprehension.*

Utilising These Notes:

- *Take an active interest in the content. Make notes, pose questions, and have peer discussions.*
- *Connect economic ideas to current affairs. Economics affects our daily lives; it is not an abstract concept. To keep your knowledge current and expanded, use additional sources such as books, articles, newspapers, and documentaries.*

Wishing you all the best in your academic endeavors!



1. INDIAN ECONOMY ON THE EVE OF INDEPENDENCE

Condition of the Indian Economy During the Pre-British Period

India's economy was independent prior to British domination. Their primary means of subsistence was agriculture. A variety of manufacturing activities were present in the economy. India's handicraft industries, which include the production of precious stone and metal works, cotton, and silk textiles, were especially well-known. These goods found a global market.

Indian economy during colonial rule

The British government's economic policies were less focused on the growth of the Indian economy and more focused on defending and advancing Britain's economic interests. It turned our economy into a buyer of British industrial goods as well as a supplier of raw materials.

There was never a serious attempt by the colonial government to calculate the national and per capita income of India. Dadabhai Naoroji, William Digby, Findlay Shirras, V.K.R.V. Rao, and R.C. Desai were made to measure India's national and per capita income during the colonial period.

V.K.R.V. Rao provided the most important estimate of national income during British rule, stating that the Indian economy grew by 2% annually. The annual growth rate of the per capita income was less than 0.5%. These two examples demonstrated how slowly the Indian economy was growing.

Condition of Agriculture Sector

The Indian economy during the British colonial era was primarily based on agriculture. Agricultural productivity stagnated during this time, primarily due to the various land and revenue settlement systems introduced by the colonial government (such as the Zamindari system). Other factors that contributed to this low productivity included inadequate technology, a lack of irrigation facilities, and fertilisers. Additionally, agricultural production became commercialised, with farmers producing cash crops instead of food crops, which the British eventually used for their own development. Lastly, inadequate flood control, drainage, low investment, etc. pulled back Indian agriculture.

Zamindari System

The British appoint zamindars as intermediaries to collect taxes from farmers. The zamindars' only concern was collecting rent; they had no regard for the farmers' financial situation. Instead of cultivating, the zamindars grab the profits made by the agricultural industry.

Condition of Industrial Sector

During colonial rule, India was unable to establish a solid industrial base. Handicraft industries in India saw a downturn. India became a large market for the British government's finished goods after the country was methodically deindustrialised, becoming only a raw material exporter. The collapse of India's domestic handicraft industries resulted in widespread unemployment.

The only significant industry founded in 1907 was the TATA Iron and Steel Company (TISCO). There were no other capital-goods industries. The GDP contribution and growth rate of the newly formed industrial

sector stayed relatively low. Railways, power generation, communications, ports, and some other departmental activities operate in the public sector.

Condition of Foreign Trade

India has been an important trading nation since ancient times. The policies of commodity production, trade, and tariffs followed by the colonial government adversely affected the structure, composition, and volume of India's foreign trade.

India became an exporter of primary products such as raw silk, cotton, wool, sugar, indigo, jute, etc. and an importer of finished consumer goods like cotton, silk, and woollen clothes and capital goods.

The British gained even more control over India's international trade with the opening of the Suez Canal. The Suez Canal is an artificial sea-level waterway in Egypt, connecting the Mediterranean Sea to the Red Sea through the Isthmus of Suez. (The Suez Canal greatly reduces the time and cost of transporting goods through waterways.)

The creation of a sizable export surplus was the primary feature of India's foreign trade during the colonial era. Gold or silver did not enter India as a result of this export surplus.

Condition of Demographic Condition

The population of British India was first collected through the census in 1881. Before 1921, India was in the first stage of demographic transition. The second stage of transition began after 1921. The year 1921 was considered the year of the "Great Divide" in Indian demographic history. The mortality rate was very high, and the infant mortality rate was quite alarming. Life expectancy was also very low.

Demographic Condition during Colonial Rule

Total literacy	16%
Female literacy	7%
Infant Mortality	238/thousand
Life Expectancy	32 years

Condition of Occupational Structure

The occupational structure means the distribution of workers across different industries and sectors. The agricultural sector accounted for the largest share of the workforce (70–75%). The manufacturing and services sectors accounted for only 10 and 15-20 percent, respectively.

Agriculture	72.56%
Industry	17.29%
Service	10.15%

Condition of Infrastructure

The British built the telegraph, ports, railroads, and waterways, among other essential infrastructure. Their colonial interests were the true driving force behind this development. The roads they constructed made it simple for the army to move supplies around India and to get raw materials to the closest port or railway station.

One of the British Empire's greatest achievements was the introduction of railways to India in 1850. In India, the railways had two effects on the economy. It made it possible for people to travel great distances. It accelerates the commercialisation of agriculture in India. India's exports increased as a result of the railways. The British also built roads, inland waterways, and sea waterways in addition to railways.

Negative Impact of British Rule

1. **Colonial Economy:** Indian economy was reduced to that of a British colony due to British political dominance.
2. **Semi Feudal Economy:** The Indian economy was a semi-feudal economy because both capitalism and feudalism were prevalent.

3. **Backward Economy:** The Indian economy degenerated as a result of the British exploitation. The characteristics that were common were extreme poverty, an unbalanced and inadequate diet, low income, caste and class oppression, unemployment, underemployment, etc.
4. **Stagnant Economy:** The economy was unable to expand and essentially remained stagnant due to the British colonial and feudal exploitation, harsh political regime, and insignificant policies.
5. **Depleted Economy:** Indian Industries gradually made efforts to satisfy the growing demand. There was severe wear and tear as well as depletion on the machinery, plants, and equipment.
6. **Fragmented Economy:** The British used a divide and conquer strategy that split India along religious, caste, and other lines.

Positive Impact of British Rule

Introduction of railway

1. Commercialisation of agriculture
2. Development of infrastructure
3. Promoted education
4. Brought certain social changes
5. Unification of India
6. Strong and efficient administration

Practice Questions

1. Analyse the impact of economic policies persuaded by the colonial government.
2. Comment on any two positive contributions made by the British in India. Discuss?
3. Critically appraise the shortfalls of the industrial policy pursued by the British colonial administration.
4. Critically appraise the shortfalls of the industrial policy pursued by the British.
5. During its colonial rule, India was unable to build a robust industrial base due to Britain's discrimination against Indian industries. Substantiate your answer.
6. During the colonial period, the Indian agricultural sector remained stagnant. Do you agree?
7. Give a quantitative appraisal of the demographic profile of India during the colonial period.
8. India became an exporter of primary products and an importer of finished goods produced in the factories of Britain. Critically examine the foreign policy adopted by the British in light of the above statement.
9. Name any two positive aspects of British rule in India.
10. Prepare a debate report on the topic: Was British rule beneficial for India?
11. Some measures taken by the British to promote their colonial interests indirectly helped India in later years. Do you agree? Justify
12. The British established some very important infrastructure for the economy, which has helped our economic development. Do you agree?
13. The British Raj had impoverished millions of people in India. Do you agree with the statement? Substantiate your answer.

14. The traditional handicrafts industries were ruined under British rule. Do you agree with this view? Justify
 15. Under colonial rule, the economic policies of the government were concerned more with the protection and promotion of the economic interests of the home country than with the development of the Indian economy. Analyse the above statement using the following indicators:
 7. a. agricultural sector; b. industrial sector; c. foreign trade; and c. infrastructure.
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2. INDIAN ECONOMY-1950-1990

Economic System:

The basic economic systems are capitalism, socialism, and mixed economies. Every economy has to answer three questions: What goods and services should be produced in the country? How should the goods and services be produced? How should goods and services be distributed among people?

In capitalism (the market economy), the forces of supply and demand answer this question. In a socialist society, the government decides what goods are to be produced in accordance with the needs of society. In mixed economies, both the government and the market together answer the three questions of what to produce, how to produce it, and how to distribute it.

Economic Planning

Economic planning is a process in which set objectives of economic development are desired to be achieved in a given period of time. A plan should have some general goals as well as specific objectives that are to be achieved within a specified period of time. In 1950, the Planning Commission was set up with the Prime Minister as its chairperson. It was renamed NITI Aayog on January 1, 2015. Planning, in the real sense of the term, began with the Second Five-Year Plan. This plan was based on the ideas of Prasanta Chandra Mahalanobis, who established the Indian Statistical Institute (ISI).

The Goals of Five-Year Plans

Growth: It refers to an increase in the country's capacity to produce the output of goods and services within the country. It implies either a larger stock of productive capital or a larger size of supporting services like transport and banking. A good indicator of economic growth is a steady increase in the gross domestic product (GDP).

GDP is the total market value of all the goods and services generated in a nation in a given year. The various economic sectors—agriculture, industry, and services—combine to form a nation's gross domestic product (GDP).

Modernisation: In order to enhance the output of goods and services, manufacturers must embrace novel technologies. The term "modernisation" refers to societal shifts as well as the application of new technologies, such as the realisation that women ought to have the same rights as men. Today's society utilises the skills of women in the workplace, in banks, in factories, in schools, etc.

Self-Reliance: A country can use resources imported from other countries or its own resources to support economic modernisation and growth. The prioritisation of self-reliance in the initial seven five-year plans entailed refraining from importing products (import substitution) that India could have produced on its own.

Equity: Every Indian should be able to meet his or her basic needs such as food, a decent house, education, and health care. Inequality in the distribution of wealth should be reduced. It is important to ensure that the benefits of economic prosperity reach to the poor sections of the society

State of Agriculture, Industry and Trade in Independent India

Agriculture

The land ceiling was another policy to promote equity in the agricultural sector. This means fixing the maximum size of land that could be owned by an individual. The big landlords challenged the legislation in the courts, delaying its implementation. In some areas, the former Zamindars continued to own large areas of land by making use of some loopholes in the legislation. Land reforms in Kerala and West Bengal were successful in implementing the policy of land to the tiller.

The Green Revolution:

At independence, about 75 percent of the country's population was dependent on agriculture. Productivity in the agricultural sector was very low because of the use of old technology and the absence of the required infrastructure. During this time, India's agriculture vitally depends on the monsoon. The stagnation in agriculture during colonial rule was permanently broken by the green revolution.

The green revolution refers to the large increase in production of food grains resulting from the use of high-yielding variety (HYV) seeds, irrigation, fertilisers, etc., especially for wheat and rice. The spread of the green revolution enabled India to achieve self-sufficiency in food grains. Due to the effects of the green revolution, a large portion of agricultural produce is sold in the market by farmers. It is called marketed surplus. The green revolution also enabled the government to procure sufficient amounts of food grains to build buffer stock that could be used in times of food shortage.

Shortcomings of Green Revolution

The shortcomings of the Green Revolution were that it would increase the disparities between small and big farmers. The HYV crops were more prone to attack by pests, and the small farmers who adopted this technology could lose everything in a pest attack. The government provided loans at a low interest rate to small farmers. As a result, the green revolution benefited small as well as rich farmers.

The Debate Over Subsidies

The economic justification of subsidies in agriculture is always debated. It is generally agreed that it is necessary to use subsidies to provide an incentive for the adoption of the new HYV technology by farmers in general and small farmers in particular. Subsidies were needed to encourage farmers to test the new technology. Economists believe that once the technology is found profitable and widely adopted, subsidies should be avoided.

Some economists believe that the government should continue agricultural subsidies because farming in India is a risky business. Most farmers are very poor, and they will not be able to afford the required inputs without subsidies. Eliminating subsidies will increase the inequality between rich and poor farmers and violate the goal of equity.

Industry and Trade

Industry provides employment that is more stable than employment in agriculture; it promotes modernisation and overall prosperity. It is for this reason that the five-year plans place a lot of emphasis on industrial development.

Industrial Policy Resolution 1956 (IPR 1956):

IPR 1956 formed the basis of the Second Five-Year Plan. This resolution classified industries into three categories. The first category comprised industries that would be exclusively owned by the state. (Strategic industries) The second category consisted of industries in which the private sector could supplement the efforts of the state sector. (The state takes the sole responsibility for starting new units.)

The third category consisted of the remaining industries, which were to be in the private sector. Although there was a category of industries left to the private sector, the sector was kept under state control through a system of licenses. (e.g., industries producing food, clothes, etc.)

Small-Scale Industry:

In 1955, the Village and Small-Scale Industries Committee, also called the Karve Committee, noted the possibility of using small-scale industries to promote rural development. A small-scale industry is defined with reference to the maximum investment allowed on the assets of a unit. At present, the maximum investment allowed is Rs. 1 crore. Small-scale industries are more 'labour intensive' as compared to large-scale industries. Therefore, it generates more employment.

Small-scale industries normally, cannot compete with big industrial firms; therefore, the production of a number of products is reserved for the small-scale industry.

Trade Policy: Import Substitution

The industrial policy that we adopted was closely related to the trade policy. In the first seven plans, trade was characterized by inward looking trade strategy. This strategy is called import substitution. It means replacing or substituting imports with domestic production. In this policy, the government protected domestic industries from foreign competition. Protection from imports took two forms: tariffs and quotas.

Tariffs are a tax on imported goods. Quotas specify the quantity of goods that can be imported. The effect of tariffs and quotas is that they restrict imports and, therefore, protect domestic firms from foreign competition.

Effect of Policies on Industrial Development

The achievements of India's industrial sector during the first seven plans are impressive. The proportion of GDP contributed by the industrial sector increased. The industrial sector became well diversified by 1990, largely due to the public sector. Protection from foreign competition enabled the development of indigenous industries.

Even though the public sector has made a substantial contribution to the growth of the Indian economy, a number of economists have voiced dissatisfaction with the performance of various public sector enterprises. Even after suffering massive losses, many public sector companies continued to operate because it is difficult to end a government project.

Import Substitution

A major thrust was given to import substitution. Domestic industries were given protection from foreign competition, and it was done through imposing heavy duty on imports. Protection from imports took two forms: tariffs and quotas. Tariffs are a tax on imported goods. Quotas specify the quantity of goods that can be imported. The effect of tariffs and quotas is that they restrict imports and, therefore, protect domestic firms from foreign competition.

Inward Looking Trade Strategy

Inward looking trade strategies were adopted as foreign trade policies. This strategy is also called import substitution. Domestic industry is offered protection from foreign

competition through import duties. Its main objective was to save foreign exchange by encouraging domestic production of goods that could be imported from the rest of the world.

Tariffs

Tariffs are taxes imposed on imported goods. They will increase the price of the goods in the domestic market. Domestic producers benefit because they receive higher prices. The government benefits by collecting Tax Revenues.

Merits of Inward-Looking Trade strategy

Rise in Industry's Share of GDP: Implementing an inward-looking trade approach can lead to an increase in the contribution of domestic industries to the overall gross domestic product (GDP).

1. **Diversification of the Industrial Sector:** By focussing on domestic production and reducing reliance on imports, countries can encourage the growth of various industries, leading to a more diversified and robust industrial sector.
2. **Investment Opportunities:** An inward-looking strategy often stimulates investment within the country. As local industries expand, there are more opportunities for both domestic and foreign investors to participate.
3. **Increase in Employment Opportunities:** Supporting domestic industries can create jobs within the country. When industries thrive, they require a skilled workforce, leading to increased employment opportunities for citizens.

Demerits of Inward-Looking Trade Strategy

1. **Inefficient Public Monopolies:** When a country focusses excessively on domestic production, it may lead to the creation of state-owned monopolies. These monopolies can become inefficient, less responsive to market demands, and hinder overall economic growth.
2. **Lack of Modernization:** Insulating the economy from international competition might discourage innovation and technological advancements. Without exposure to global best practices, industries may lag behind in terms of modernisation and efficiency.
3. **Concentration of power in Public Sectors:** An inward-looking approach often involves heavy government intervention. This concentration of power in the public sector can lead to bureaucracy, corruption, and inefficiencies.
4. **Non-Viable Government Enterprises:** Protecting domestic industries can sometimes result in the survival of uncompetitive government-owned enterprises. These enterprises may drain resources without contributing significantly to economic development.

Role of Public Sector in India

1. **Income Generation:** Public-sector enterprises contribute significantly to the national income. They engage in various economic activities, from manufacturing to service provision, which collectively add to the country's GDP.
2. **Capital Formation:** The public sector is instrumental in generating capital within the Indian economy. It invests in infrastructure projects, industries, and other development initiatives, thereby contributing to the overall capital formation.
3. **Employment Creation:** Public sector undertakings (PSUs) employ millions of people across different sectors. By providing stable jobs, they contribute to reducing unemployment and improving livelihoods.
4. **Infrastructure Development:** Public sector organisations invest in critical infrastructure such as roads, railways, power plants, and telecommunications. These developments enhance connectivity, facilitate economic activities, and improve the quality of life for citizens.
5. **Strong Industrial Base:** PSUs have played a pivotal role in establishing and nurturing India's industrial base. They set up factories, manufacturing units, and research facilities, contributing to the growth of various sectors.

6. **Export Promotion and Import Substitution:** Public-sector enterprises actively participate in export-orientated activities. They promote exports by manufacturing goods for international markets. Simultaneously, they also work towards import substitution—producing domestically what was previously imported.
7. **Contribution to Central Exchequer:** Through taxes, dividends, and other revenue-sharing mechanisms, PSUs contribute to the government's coffers. This revenue supports public welfare programmes and developmental initiatives.
8. **Checking Concentration of Income and Wealth:** By ensuring equitable distribution of resources and opportunities, the public sector helps prevent extreme concentration of wealth and promotes social justice.
9. **Removal of Regional Disparities:** Public sector investments are often directed towards less-developed regions. Establishing industries and infrastructure in these areas helps bridge regional disparities and promotes balanced growth.

Features of Economic Planning till 1991

1. Dependence on the public sector
2. Controlled development of the private sector
3. Inward-looking trade strategy
4. Protection of small-scale industries
5. Diversification of industries
6. Self-sufficiency in food grains

Achievements of Planning till 1991

1. **Increase in National Product or National Income:** India's planning efforts contributed significantly to the growth of our national income. Over the decades, we witnessed substantial expansion in various sectors, from agriculture to industry and services. The Five-Year Plans played a crucial role in shaping modern India.
2. **Increase in Per Capita Income:** Per capita income is the measure of prosperity that reflects how well economic resources are distributed among citizens. Though India's overall income increased, it is essential to recognise that equitable distribution remained a challenge. The planning process did manage to lift many out of poverty and improve living standards.
3. **Generation of Employment:** Employment generation is arguably one of the most critical aspects of economic planning. The plans aimed at creating jobs across sectors, especially in agriculture, industry, and services.
4. **Diversification of Industrial Sectors:** India's planning emphasised moving away from an agrarian-centric economy. We saw the rise of manufacturing, textiles, heavy industries, and even the tech sector.
5. **Development of Economic and Social Infrastructure:** Infrastructure is the backbone of progress. Roads, railways, power plants, schools, hospitals, etc. are examples of economic and social infrastructure. The five-year plans laid the foundations of this economic and social infrastructure.
6. **Rise in Capital Formation:** The growth engine is capital formation. The programmes promoted investing and saving, which resulted in the capital accumulation needed to strengthen the economy.

Failure of Planning till 1991

1. Increase in Poverty
2. Consistent Rise in Prices
3. Problems of unemployment
4. Inefficient Public Sector Monopolies
5. Concentration of Economic Power
6. Increased Corruption

Practice Questions

1. 'Food security and the Green Revolution' Write any four arguments in favour of the Green Revolution.
2. Discuss the Green Revolution and its consequences
3. Do you think import substitution can protect domestic industries? Give your views.
4. Explain 'growth with equity' as a planning objective.
5. Explain the need for and types of land reforms implemented in the agriculture sector.
6. Five-year plans play a commanding role in the economic development of India. Mention the objectives and achievements of India's five-year plans.
7. India started five-year plans as a strategy of development in 1950. Since then, we have completed 11 five-year plans. Draw a time line for our five-year plans.
8. List out some arguments in favour of and against the Green Revolution.
9. Modernisation means the adoption of new technology and a change in the social outlook of the people, which is one of the most important objectives of Indian planning. What are the other objectives?
10. Prepare an essay on the 'Goals of Five -Year Plans' pursued until 1991.
11. Self-reliance was an important goal of the five-year plan in India. Give reasons
12. The Green Revolution brought about self-sufficiency in food grain production. Do you agree?
13. The Green Revolution caused the degradation of soil due to heavy doses of chemical fertilisers and pesticides. Substantiate.
14. The Green Revolution enabled India to achieve self-sufficiency in food grains. How far is this statement true? Substantiate your answer.
15. The Green Revolution enabled India to achieve self-sufficiency in food grains. Substantiate your answer.
16. The spread of green revolution technology enabled India to achieve self-sufficiency in food grains." In light of the above statement, explain the advantages of the green revolution.
17. The stagnation in Indian agriculture during colonial rule was permanently broken by the Green Revolution. Comment on the statement.
18. What are miracle seeds, and how did they impact agriculture?
19. What are the common objectives of five-year plans? Do you think that we achieved these objectives with our five decades of planning?
20. What is a marketable surplus in agriculture?
21. What is a plan, and why did India opt for planning?
22. Why was it necessary for a developing country like India to follow self-reliance as a planning objective?

3. ECONOMIC REFORMS SINCE 1991

Liberalisation, Privatisation and Globalisation:

The origin of the financial crisis can be traced to the inefficient management of the Indian economy in the 1980s. Expenditure is more than income; the government borrows to finance the deficit from banks and also from people within the country and from international financial institutions.

The government was spending a large share of its income on areas such as the social sector and defence, which do not provide immediate returns. The income from public sector

undertakings was also not very high to meet the growing expenditure. Foreign exchange, borrowed from other countries and international financial institutions, was spent on meeting consumption needs. In the late 1980s, government expenditure began to exceed its revenue by large margins. No country or international funder was willing to lend to India.

India approached the International Bank for Reconstruction and Development (IBRD), popularly known as the World Bank and the International Monetary Fund (IMF), and received \$7 billion as a loan to manage the crisis. (*The International Bank for Reconstruction and Development (IBRD) and International Monetary Fund (IMF) were established by delegates at the Bretton Woods Conference in 1944 and became operational in 1946.*)

There were two components to the New Economic Policy (NEP): the stabilisation measures and the structural reform measures.

Stabilisation measures are short-term measures, intended to correct some of the weaknesses in the balance of payments and to bring inflation under control.

Structural reform policies are long-term measures aimed at improving the efficiency of the economy and increasing its international competitiveness by removing the rigidities in various segments of the Indian economy.

Liberalisation

Liberalisation was introduced to put an end to the restrictions and open various sectors of the economy. Sectors that received greater attention in and after 1991 are the industrial sector, financial sector, tax reforms, foreign exchange markets, trade, and investment.

Deregulation of Industrial Sector

In India, regulatory mechanisms were enforced in various ways: (I) industrial licensing; (II) the private sector was not allowed in many industries; (III) some goods could be produced only in small-scale industries; and (IV) controls on price fixation and distribution of selected industrial products.

The reform policies introduced in and after 1991 removed so many restrictions. Industrial licensing was abolished for almost all products. (Except for alcohol, cigarettes, hazardous chemicals, etc.), the only industries that are now reserved for the public sector are defence equipment, atomic energy generation, and railway transport. Many goods produced by small-scale industries have now gone unreserved. In many industries, the market has been allowed to determine the prices.

Financial Sector Reforms

The financial sector includes financial institutions, such as commercial banks, investment banks, stock exchange operations, and foreign exchange markets. The Reserve Bank of India (RBI) oversees the regulation of the financial industry in India. One of the major aims of financial sector reforms is to reduce the role of the RBI from regulator to facilitator of the financial sector.

The reform policies led to the establishment of private sector banks, both Indian and foreign. Foreign institutional investors (FII), such as merchant bankers, mutual funds, and pension funds, are now allowed to invest in Indian financial markets. The foreign investment limit in banks was raised to around 50 percent. Those banks that fulfil certain conditions have been given the freedom to set up new branches without the approval of the RBI.

Tax Reforms

Tax reforms are concerned with the reforms in the government's taxation and public expenditure policies, which are collectively known as fiscal policy. There are two types of taxes: direct and indirect. Direct taxes consist of taxes on the incomes of individuals as well as

the profits of business enterprises. Indirect taxes are taxes levied on commodities. The rate of corporation tax, which was very high earlier, has been gradually reduced. Recently, the Parliament passed a law, the Goods and Services Tax Act 2016, to simplify and introduce a unified indirect tax system in India. This law came into effect in July 2017. GST is expected to generate additional revenue for the government, reduce tax evasion, and create 'one nation, one tax, and one market.'

Goods and Service Tax (GST)

Goods and Service Tax (GST) is the single comprehensive indirect tax, operational from July 1, 2017, on the supply of goods and services, right from the manufacturer or service provider to the consumer. It is applicable throughout the country with one rate for one type of goods or service. It has amalgamated a large number of central and state taxes and cesses. It has replaced a large number of taxes on goods and services levied on the production, sale, or provision of services.

It has replaced various types of taxes/cesses, levied by the central and state/UT governments. Some of the major taxes that were levied by the Centre were Central Excise Duty, Service Tax, Central Sales Tax, Cesses, Taxes on Advertisements, Taxes on Lottery, Betting, Gambling, State Cesses on Goods, etc. These have been subsumed in GST. Five petroleum products have been kept out of GST for the time being, but with the passage of time, they will become subject to GST. State governments will continue to levy VAT on alcoholic liquor for human consumption. Tobacco and tobacco products will attract both GST and Central Excise Duty.

Under GST, there are six (six) standard rates applied, i.e., 0%, 5%, 12%, 18%, and 28%, on the supply of all goods and/or services across the country.

Salient Features of GST

1. **One Nation, One Tax:** The Central and State Governments imposed the Goods and Services Tax (GST), which replaced a number of indirect taxes, such as value-added tax (VAT), excise duty, and service tax. It brought uniformity to India's tax system and removed the cascading effect of taxes.
2. **Dual Structure:** The Central Government's (CGST) and the State Governments' (SGST) levies are the two different taxation regimes that make up the GST.
3. **Destination-based Tax:** Every transaction in the supply chain, from the producer to the final consumer, is subject to the destination-based tax known as GST.
4. **Input Tax Credit (ITC):** Businesses can claim a credit for the taxes they have paid on inputs used in the production or provision of goods and services under the GST by using the input tax credit. This helps avoid double taxation in addition to reducing the overall tax liability.
5. Except alcohol, all goods sold for human consumption would be come under GST.
6. **Increased Compliance and Transparency:** Through the expansion of the formal economy, GST seeks to improve tax compliance. The electronic records and digitisation of processes within the tax system contribute to the transparency of the system and reduce instances of tax evasion.

Foreign Exchange Reforms

The first important reform in the external sector was made in the foreign exchange market. In 1991, as an immediate measure to resolve the balance of payments crisis, the rupee was devalued (a decrease in the value of home currency in the international market due to deliberate action taken by the government) against foreign currencies.

It also set the tone to free the determination of rupee value in the foreign exchange market from government control. It led to an increase in the inflow of foreign exchange. Now, more often than not, markets determine exchange rates based on the demand and supply of foreign exchange.

Trade and Investment Policy Reforms:

The liberalisation of trade and investment policy was initiated to increase the international competitiveness of industrial production as well as foreign investments and technology in the economy. The aim was also to promote the efficiency of local industries and the adoption of modern technologies. In order to protect domestic industries, India was following a regime of quantitative restrictions on imports.

The trade policy reforms aimed at (i) the dismantling of quantitative restrictions on imports and exports; (ii) the reduction of tariff rates; and (iii) the removal of licensing procedures for imports. Import licensing was abolished except in the case of hazardous and environmentally sensitive industries.

Quantitative restrictions on imports of manufactured consumer goods and agricultural products were also fully removed in April 2001. Export duties have been removed to increase the competitive position of Indian goods in international markets.

Privatisation

It implies the shedding of ownership or management of a government-owned enterprise. Government companies are converted into private companies in two ways: (i) by the withdrawal of the government from ownership and management of public sector companies; and (ii) by the outright sale of public sector companies.

Privatisation of public-sector enterprises by selling off part of the equity of PSEs to the public is known as disinvestment. Some examples of public enterprises with Maharatnas, Navratnas, and Miniratnas status are:

1. **Maharatnas:** (a) Indian Oil Corporation Limited, and (b) Steel Authority of India Limited,
2. **Navratnas:** (a) Hindustan Aeronautics Limited, (b) Mahanagar Telephone Nigam Limited
3. **Miniratnas:** (a) Bharat Sanchar Nigam Limited; (b) Airport Authority of India; and (c) Indian Railway Catering and Tourism Corporation Limited.

Globalisation

Globalisation generally refers to the integration of the economy of a country with the world economy. It involves the creation of networks and activities that eliminate economic, social, and geographical boundaries.

Outsourcing

Outsourcing is one of the important outcomes of the globalisation process. In outsourcing, a company hires regular services from external sources. As a form of economic activity, outsourcing has widely expanded. Many of the services, such as voice-based business processes (popularly known as BPO or call centres), record keeping, accounting, banking services, music recording, film editing, book transcription, clinical advice, etc., are outsourced by companies in developed countries to India. The low wage rates and availability of skilled workforce in India have made it a destination for global outsourcing in the post-reform period.

World Trade Organisation (WTO):

The WTO was founded in 1995 as the successor organization to the General Agreement on Trade and Tariff (GATT).

GATT was established in 1948. The WTO is expected to establish a rule-based trading regime in which nations cannot place arbitrary restrictions on trade. The purpose of the WTO is to increase production and trade of services, ensure optimum utilisation of world resources, and protect the environment. The WTO agreements cover trade in goods as well as services to facilitate international trade (bilateral and multilateral) through the removal of tariffs as well as non-tariff barriers between member countries. As an important member of the WTO, India has

been at the forefront of framing fair global rules and regulations and advocating for the interests of the developing world.

Indian Economy During Reforms: An Assessment

In economics, the growth of an economy is measured by the gross domestic product (GDP). In the post-1991 period, India witnessed rapid growth in GDP on a continuous basis for decades. During the reform period, the growth of agriculture has declined. While the industrial sector was in fluctuation, the growth of the service sector has improved. The opening of the economy has led to a rapid increase in foreign direct investment and foreign exchange reserves. The foreign investment includes foreign direct investment (FDI) and foreign institutional investment (FII). India became successful in exporting auto parts, engineering goods, IT software, and textiles during the reform period. Rising prices have also been kept under control.

Reforms in various sectors

The reform process has been widely criticised because it does not fully address the areas of employment, agriculture, industry, infrastructure development, and fiscal management.

Growth and Employment:

Though the GDP growth rate has increased during the reform period, scholars point out that the reform-led growth has not generated sufficient employment opportunities in the country.

Reforms in Agriculture:

Reforms do not benefit agriculture. Public investment in the agriculture sector, especially in infrastructure, has fallen during the reform period. The removal of fertiliser subsidies has led to an increase in the cost of production, which has severely affected small and marginal farmers. Export-orientated policy strategies in agriculture lead to focussing on cash crops. This leads to an increase in the price of food grains.

Reforms in Industry:

Industrial growth has also recorded a slowdown. This is because of decreasing demand for industrial products due to cheaper imports, inadequate investment in infrastructure, etc. The infrastructure facilities, including power supply, have remained inadequate due to a lack of investment. Developing countries like India still do not have access to developed countries' markets because of high non-tariff barriers.

Disinvestment Policy:

Every year, the government fixes a target for disinvestment in public sector enterprises. Critics point out that the assets of public-sector enterprises have been undervalued and sold to the private sector. Disinvestment is used to offset the shortage of government revenues rather than develop social infrastructure.

Reforms and Fiscal Policies:

The tax reductions in the reform period were aimed at yielding larger revenue and curbing tax evasion, but they did not achieve a fruitful result. Reform policies involving tariff reduction have curtailed the scope for raising revenue through customs duties.

Practice Questions

1. What were the objectives of NEP 1991?
2. Critically evaluate the impact of globalisation on the agricultural and industrial sectors in India.
3. Critically evaluate the impact of the New Economic Policy of 1991.

4. Describe the circumstances that led to the introduction of new economic reforms in the Indian economy.
5. Do you agree with the privatisation of profit-making public enterprises? Substantiate your answer.
6. Do you think outsourcing is good for India? Substantiate your views.
7. Does the New Economic Policy of 1991 adversely affect agricultural development in India? Provide evidence to support your response.
8. Evaluate the privatisation of the Indian economy in the light of social justice.
9. Explain globalisation in the context of the Indian economy.
10. Liberalisation involves liberating the economy from control. Discuss the process of liberalisation introduced in India since 1991.
11. List out some arguments against the New Economic Policy of 1991.
12. List out three points each in favour of and against privatisation.
13. Prepare an essay on privatisation in the Indian economy.
14. The Indian economy's small-scale industries are not strong enough to withstand global competition. In light of recent economic policies, do you agree with this viewpoint?
15. What are your views on privatization? What is the impact of privatisation on telecom, banking, and insurance? Substantiate your answer.
16. What is meant by disinvestment?
17. Write an essay on globalisation and its implications for India.
18. Write a paper on the New Economic Policy, which the Indian government implemented in 1990–1991 as a long-term plan for the country's development. What is meant by disinvestment?
19. (Hint: globalisation, privatisation, and liberalisation.)
20. Write a description outlining the advantages and disadvantages of the 1991 New Economic Policy.
21. The economic reforms of 1991 have been implemented in a crisis of foreign exchange reserves. Identify the reform measures adopted by the government of India to overcome the crisis.
22. Discuss the various reforms in the financial sector in NEP 1991.
23. What were the tax reforms implemented in India in connection with NEP 1991?
24. What were the reasons for introducing the economic reforms in 1991?
 - a. The Gulf war
 - b. The negative balance of payments
 - c. Increase in fiscal deficit
 - d. All of the above
25. Match the following:

A	B	C
Liberalisation	Disinvestment	Loss making PSUs
Privatization	Outsourcing	Skilled and cheap man power
Globalisation	Abolished licensing	Tax Reforms

4. POVERTY

Poverty

Poverty refers to a state in which an individual is unable to fulfil even the basic necessities of life. The minimum requirement includes food, clothing, shelter, and health facilities.

Relative Poverty

Absolute poverty refers to a phenomenon in which a section of society is unable to fulfil even its basic necessities of life concerning food, clothing, housing, education, and health.

Absolute Poverty

Relative poverty refers to a phenomenon in which inequalities of income exist and, hence, one person is poor in relation to another.

Concept of Poverty Line

The poverty line is a cut-off point on the line of distribution, which usually divides the population of a country into poor and non-poor groups. People with incomes below the poverty line are called poor (BPL), and people with incomes above the poverty line are called non poor (APL).

The Y K Alagh Committee (1979) defined the poverty line on the basis of a recommended nutritional requirement of 2400 calories per person in rural areas and 2100 for a person in urban areas.

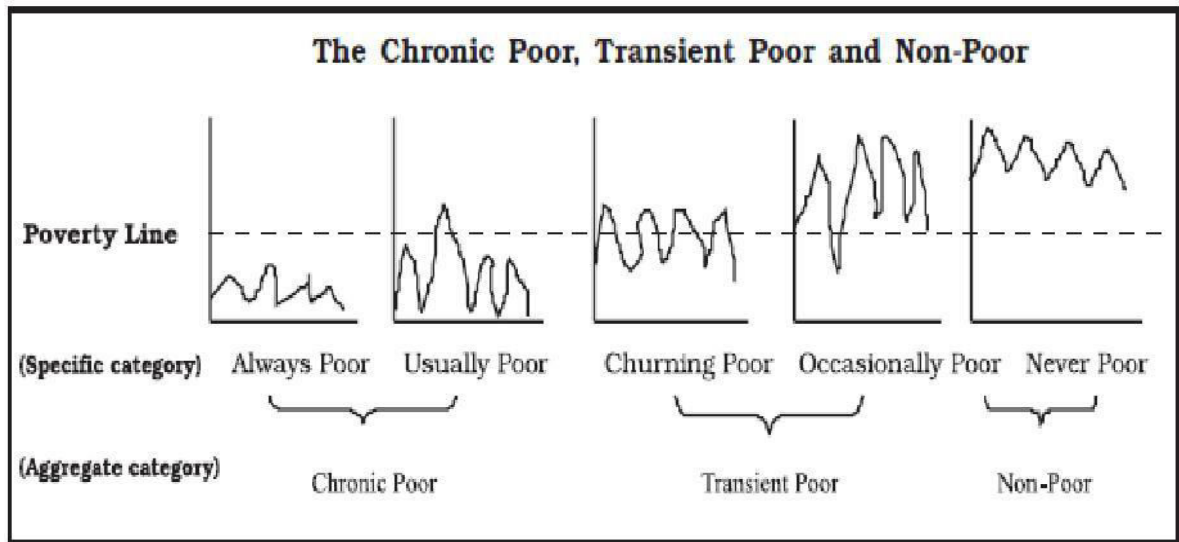
1. In rupee terms, at 2000-2001 prices, the consumption worth of ₹328 per person a month in rural areas and ₹454 for urban areas was also considered for the poverty line.
2. At 2005-2006 prices, the consumption worth of ₹368 per person a month in rural areas and ₹558 in urban areas was also considered for the poverty line.
3. In 2011, the Suresh Tendulkar Committee adopted the cost of living as the basis for identifying poverty. The panel stipulated a benchmark daily per capita expenditure of ₹ 27 and ₹ 33 in rural and urban areas, respectively.
4. As per the report submitted by the C. Rangarajan Committee (2014) to the Government of India, those spending over ₹ 32 a day in rural areas and ₹ 47 in towns and cities should not be considered poor. The Rangarajan Committee estimates that 3 out of 10 Indians are poor.

A person is considered poor if his or her income or consumption level falls below a given minimum level necessary to fulfil basic needs. Each country uses an imaginary line that is considered appropriate for its existing level of development and its accepted minimum social norms. This is called the poverty line.

In simple terms, the poverty line is the monetary income that one must have to afford the basic amenities of life. Globally, the poverty line is set at \$2.15 per day. This amount was recently updated by the World Bank for 2022. Depending upon the country's social and economic status required to attain its basic needs, the poverty line may vary from one country to another. In India, the poverty line lies at 1,286 rupees per month for urban areas and 1,059.42 rupees per month for rural areas.

Categorizing Poverty

1. **Chronic poor:** It includes people who are always poor and those who are usually poor.
2. **Transient Poor:** It includes those who are Churning Poor (who regularly move in and out of poverty, like small farmers) and Occasionally Poor (who are rich most of the time and poor sometimes).
3. **Non-poor:** They are never poor.



Characters of the Poor

1. Hunger, Starvation and Malnutrition
2. Poor Health
3. Poor Educational Achievement
4. Limited Economic Opportunities
5. Debt Trap
6. Lack of Facilities of Electricity and Water
7. Bigger Family Size

Criticism of the Poverty Line

This method does not differentiate between the very poor and the other poor. This method ignores factors like accessibility to basic education, health care, drinking water, etc. This method ignores social factors that generate and are responsible for poverty, like illiteracy, ill health, and a lack of access to resources. This method ignores civil and political freedom.

Tools to Measure Poverty

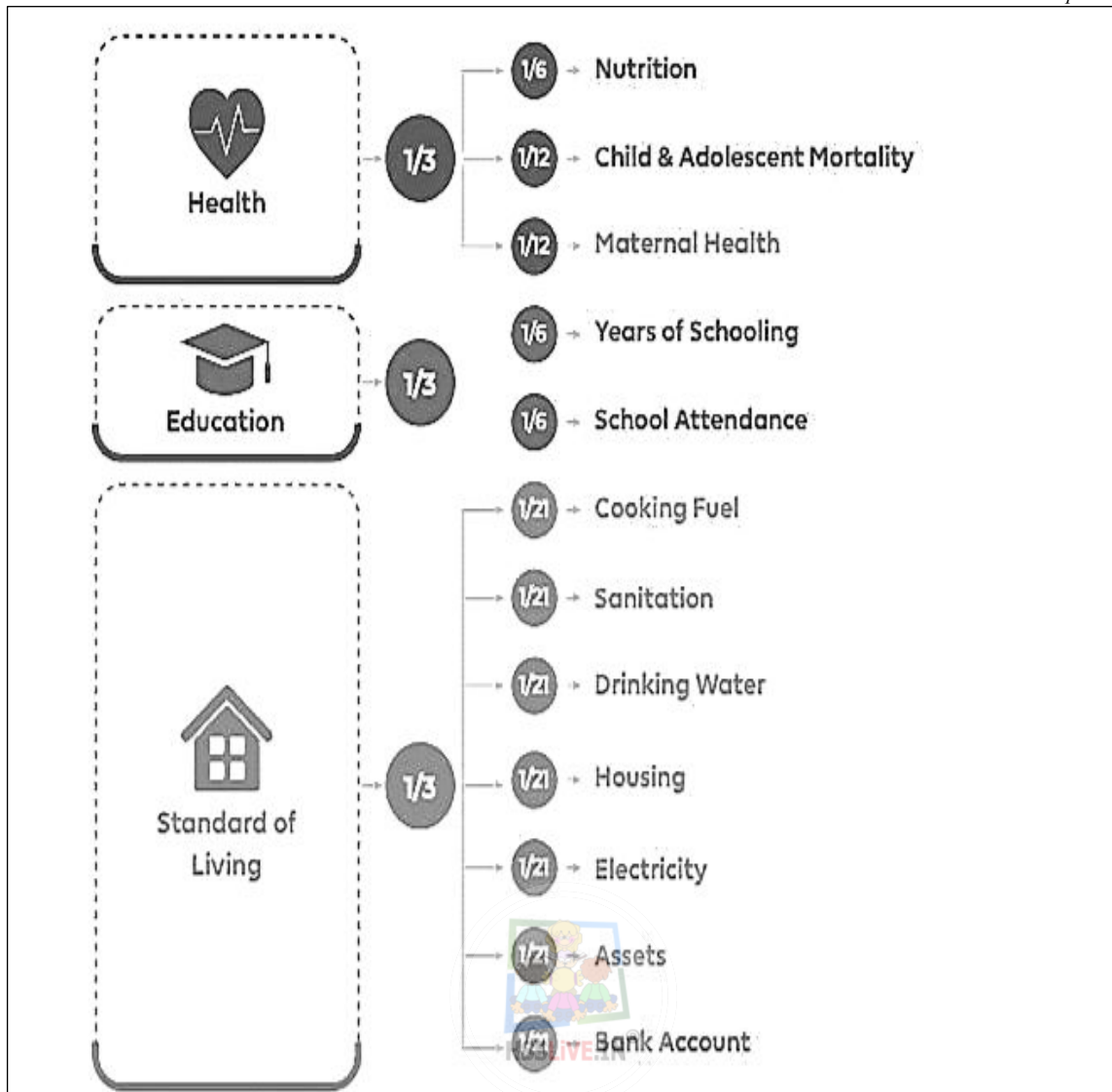
1. The simplest method to determine the number of poor is the headcount ratio (HCR). The head count ratio is calculated by dividing the number of people below the poverty line by the total population.
2. Amartya Sen a Noble Laureate developed “Sen Index to measure poverty.”
3. Other tools to measure poverty are the Gap Index and the Squared Poverty Gap.
4. Official data on poverty is collected by the NSSO.

National Multidimensional Poverty Index

In the past, income has been the primary determinant of poverty in most estimations. On the other hand, the Alkire-Foster (AF) methodology-based Global Multidimensional Poverty Index (MPI) records overlapping deprivations in living standards, health, and education. Due to its direct measurement and comparison of deprivations, it is a complement to income poverty measurements.

The Oxford Poverty and Human Development Initiative (OPHI) and the United Nations Development Programme (UNDP) collaborate to publish the global MPI Report.

The three equally weighted dimensions of India's national MPI—standard of living, education, and health—are each represented by twelve indicators. The graphic below illustrates these:



Causes of Poverty

1. Underdevelopment of the Indian Economy:

Over the past few decades, India's economy has grown significantly, but sadly, not everyone has benefited equally from this progress. Poverty is sustained by uneven development across areas and industries. Certain places are still undeveloped, devoid of opportunities and basic infrastructure.

2. Population Explosion:

The enormous population of India presents both opportunities and difficulties. Infrastructure, social services, and resources are all strained by the rapid population growth. It becomes increasingly difficult to provide for the needs of such a large population.

3. High Level of Unemployment:

Unemployment is still a problem even in the face of economic expansion. Particularly for young people, there are often more job openings than there are job seekers. Poverty is also a result of underemployment, which occurs when people work in jobs that do not fully utilise their skills.

4. Inequalities of Income:

India has extreme income inequality. Many people in the population struggle to make ends meet, while a small percentage enjoy substantial wealth. Poverty is made worse by the concentration of wealth in a small number of people.

5. **Inflation:**

The cost of necessities is rising, which disproportionately impacts low-income households. Basic necessities become more difficult for families to afford when inflation outpaces income growth.

6. **High Illiteracy Rate:**

Breaking the cycle of poverty can be effectively achieved through education. However social mobility is hampered by a lack of access to high-quality education, particularly in rural areas. Poverty is sustained by illiteracy because it reduces employment opportunities.

7. **Poor State of Agriculture:**

In India, agriculture continues to be an important industry that employs a sizable percentage of the workforce. However, low agricultural productivity is a result of antiquated farming methods, inadequate irrigation, and erratic weather. Farmers frequently battle poverty and debt.

8. **Underutilization of Resources:**

Natural resources abound in India, but they are not always used effectively. The best use of resources—be they minerals, water, or land—is impeded by poor infrastructure and management.

9. **Social Factors:**

India's social structure is very important. Obstacles include the caste system, historical injustices, and discrimination against women, people of colour, or people of religion. Certain populations are systematically disadvantaged when it comes to work, healthcare, and education.

10. **Political Factors:**

Poverty is impacted by political choices. The environment is shaped by laws pertaining to resource distribution, social welfare, and economic development.

How to Remove Poverty

1. Acceleration of Economic Growth
2. Reducing Inequalities of Income
3. Population Control
4. Agricultural Development
5. More Employment Opportunities
6. Land Reforms

Three Dimensional Approaches to Remove Poverty

1. **Growth Oriented Approach (Trickle Down)**

The First approach is based on assumption that rapid increase in GDP & per capital income would help in removal of poverty

2. **Poverty Alleviation Programme**

The second approach is that government has introduce a variety of programme for eradication of poverty.

3. Minimum Needs Programme

The third approach is to provide minimum basic amenities to the people. For example, Food grains at subsidized rates, education, health, water supply & sanitation will improve standard of living people

Policies and Programme towards Poverty Alleviations

1. Acceleration of Economic Growth
2. Control over Population
3. Create More Employment Opportunity
4. Accelerate Land Reforms
5. Maintain Need Programmes for the Poor
6. Reduction in Economic Inequalities
7. Rapid Rural Industrialization
8. Control over Inflation
9. Accelerate Social Security

Poverty Alleviation Programme in India

1. Annapurna
2. Antyodaya
3. Integrated Rural Development Programme (IRDP)
4. Jawahar Gram Samridhi Yojana (JGSY)
5. National Family Benefit Scheme (NFBS)
6. National Maternity Benefit Scheme
7. National Old Age Pension Scheme (NOAPS)
8. National Rural Employment Guarantee Act (NREGA)
9. Pradhan Mantri Gramin Awaas Yojana.

Measures to Accelerate Rate of Economic Growth

1. Fuller and more productive employment for the poor
2. Minimum and fair wages for poor
3. Increase in self-employment opportunities
4. Provision for education, training, and health to improve labour productivity
5. Establishment of cottage and small-scale industries in backward rural regions of the country
6. Efficient use of natural, human, and capital resources of the country

Poverty Alleviation Programme a Critical Assessment

1. Poverty Alleviation Programme is successful to great extent since independence.
2. The percentage of absolute poor in some states is now well below the national average.
3. Despite various strategies to alleviate poverty, hunger, malnourishment, illiteracy and lack of basic amenities Unequal distribution of land is still exist
4. Resources and fund allocation is still insufficient
5. Lack of commitment and corruption prevailing in the bank and government officials
6. Lack of information provided to the poor regarding various schemes

Poverty Alleviation Schemes

1. **The Integrated Rural Development Programme (IRDP):** The primary goal of the IRDP is to end poverty, hunger, and unemployment in rural India. It was first introduced in 1978–1979 and went nationwide on October 2, 1980. The programme was designed to assist the impoverished in rural areas by offering bank loans and opportunities for employment subsidies.
2. **Jawahar Rozgar Yojana/Jawahar Gram Samridhi Yojana (JRY):** The goal of this programme, which was launched on April 1, 1999, is to make a better life for the impoverished in rural areas. The JRY sought to give the unemployed and underemployed in rural areas meaningful work opportunities through the development of economic infrastructure,

community resources, and social assets. The Jawahar Gramme Samridhi Yojana is being executed in its entirety at the gramma panchayat level.

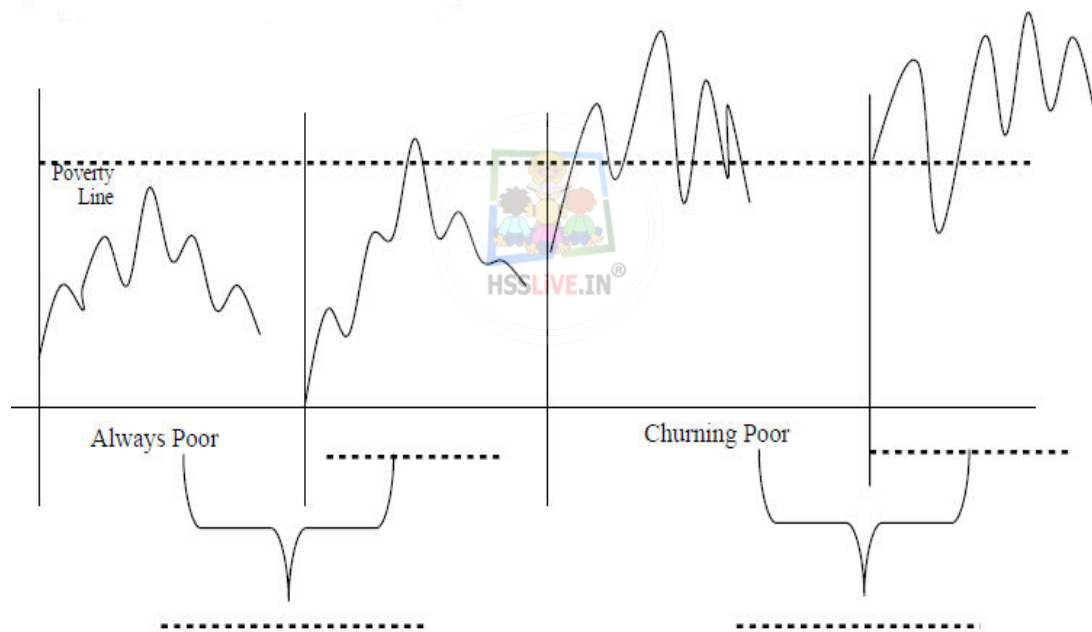
3. **Indira Awaas Yojana for Rural Housing:** The goal of the Indira Awaas Yojana (LAY) programme is to give free housing to families belonging to the SC/ST below the poverty line (BPL) in rural areas. Launched in 2015 as Housing for All by 2022, the programme is comparable to the Indira Awas Yojana for the urban poor.
4. **Food for Work Programme:** It is one of the most important poverty alleviation programme in India. The Food for Work Programme was founded with the cooperation of the Ministry of Rural Development and state governments. It seeks to improve food security by increasing wage employment. All rural residents who need to work for a living and desire to perform manual, unskilled labour are eligible for the programme.
5. **National Old Age Pension Scheme:** The government of India offers this pension. Panchayats and municipalities are entrusted with carrying out this scheme across the states and union territories. Depending on the state, the state contribution may change.
6. **Sampoorna Gramin Rozgar Yojana:** It is one of the most important policies for poverty alleviation. The Sampoorna Gramin Rozgar Yojana, which combined the provisions of the Jawahar Gram Samridhi Yojana and the Employment Assurance Scheme (EAS), was formed on September 25, 2001. (JGSY). The initiative was to be put into action through the Panchayati Raj institutions (PRIs). The main objectives of the programme are to provide wage jobs, build durable economic infrastructure in rural areas, and ensure food and nutrition security for the poor.
7. **Mahatma Gandhi National Rural Employment Guarantee Act of 2005 (MGNREGA):** The Act provides 100 days of guaranteed employment per year to every rural household. Only one-third of the positions would be open to women. The Government of India will also create National Employment Guarantee Funds. To implement the programme, state governments will also establish State Employment Guarantee Funds.
8. **Aajeevika (2011):** National Rural Livelihood Mission It aims to diversify the needs of the impoverished in rural areas and give them jobs with steady monthly wages. At the village level, self-help groups are formed to assist the poor.
9. **The National Urban Livelihood Mission (NULM):** It is one of the most important poverty-alleviation programmes in India. It organises urban poor people into self-help groups, provides chances for skill development that lead to market-based employment, and assists them in starting their own businesses by ensuring easy access to credit.
10. **Pradhan Mantri Kaushal Vikas Yojana:** The focus of this programme to reduce poverty will be on people who are just starting their careers, especially dropouts from classes X and XII.
11. **Pradhan Mantri Jan Dhan Yojana:** Its objective was the direct benefit transfer of subsidies, pensions, insurance, and other benefits. The programme's main target group is the unbanked poor.
12. **National Food Security Act (NFSA):** The 2013 National Food Security Act (NFSA) now governs the Public Distribution System (PDS). This law covers the Public Distribution System, the Integrated Child Development Services Programme, and the Midday Meal Programme. The NFSA 2013 also acknowledges maternity benefits.
13. **Gram Swaraj Abhiyan (GSA):** The main focus of this scheme is "Sabka Sath, Sabka Gaon, Sabka Vikas." Its main objectives are to advance social harmony, increase public knowledge of government programmes that help the poor.
14. **Pradhan Mantri Adarsh Gram Yojana (PMAGY):** This poverty alleviation programme aims to integrate the development of specific villages with more than 50% Scheduled Caste (SC) population by implementing current schemes of the Central and State Governments in a convergent manner and making use of gap-filling funds provided by Central Assistance.
15. **Sansad Adarsh Gram Yojana (SAGY)** Under the Sansad Aadarsh Gram Yojana (SAGY), each Member of Parliament (MP) is tasked with improving the physical and socio-economic

infrastructure in three villages by 2019. By 2024, the plan aims to build a total of eight “Adarsh Villages” or “Model Villages.”

16. **National Social Assistance Programme:** The Ministry of Rural Development is in charge of the National Social Assistance Programme. In providing social assistance benefits to poor households in the case of old age, death of the breadwinner and maternity, the NSAP aims at ensuring minimum national standards, in addition to the benefits that the States are currently providing or might provide in future

Practice Questions

- Who is a poor man?
- Distinguish between absolute poverty and relative poverty.
- What are the various causes of poverty?
- What do you understand by poverty? How is poverty line fixed and poverty measured?
- List out the major poverty alleviation programmes implemented in India under the following headings: Self-employment, wage -employment, social security, food security.
- Fill up appropriately.



- Do you think that the poverty alleviation programmes adopted in India achieved their objectives? Justify your answer.
- Categorise different types of poverty.
- Imagine you are assigned to the position of village officer. Make appropriate recommendations for addressing the issues related to poverty?
- Do you think poverty and unemployment are mutually reinforcing? Provide evidence.
- Briefly explain the three-dimensional approaches to poverty adopted by the government.
- Prepare an essay on the topic ‘Poverty alleviation programme.’

5. HUMAN CAPITAL FORMATION IN INDIA

Various Resources

Capital

Capital resources are goods made and used to produce other goods and services. Examples include buildings, machinery, tools, and equipment. Capital resources help improve productivity. Capital resources are also called capital goods, or simply capital, or physical capital.

Physical Capital

Physical capital refers to all those inputs that are required for further production, like plants and machinery, factories, buildings, raw materials, etc.

Human Capital

The term human capital refers to the economic value of a worker's experience and skills. Human capital includes assets like education, training, intelligence, skills, health, and other things employers value, such as loyalty and punctuality.

Human Capital Formation

Human capital formation means the development of abilities and skills among the population of a country.

<i>PHYSICAL CAPITAL</i>	HUMAN CAPITAL
Physical capital is tangible (touchable)	Human capital is intangible (untouchable)
It can be easily sold in the market	It cannot be sold in the market.
It depreciates with the passage of time	Depreciation in human capital can be reduced by making continuous investment in education & health
It is more mobile between countries Physical capital (like machinery) separated from its owners.	Human capital is less mobile between countries as compared to physical capital.
Physical capital is the outcome of the conscious decision of the owner and is mainly an economic and technical process.	Formation of human capital is partly a social process by partly a conscious decision of the possessor of the human capital.

Importance of Human capital Formation

1. Effective Use of Physical Capital.
2. Effective Use of Human Capital
3. Increases Life Expectancy
4. Improves Quality of Life
5. Control of Population Growth

Human Capital and Economic Growth:

A country's real national income rising indicates economic growth. Naturally, the contribution of the educated person to economic growth is greater than that of the illiterate person. A healthy person could provide labour supply for a longer period of time; therefore, health is also an important factor for economic growth. Education and health, job training, job market information, and migration increase the income-generating capacity of an individual.

Education provides knowledge to understand changes in society and scientific advancements, which helps invention and innovation. Thus, economic growth in each sector has reinforced the growth of every other sector.

Reports on the Indian economy in recent times have identified that India would grow faster due to its strength in human capital formation. Human capital treats human beings as a means to an end, the end being the increase in productivity. Human welfare should be increased through investments in education and health.

Sources of Human Capital

1. Investment in Education
2. Investment in health

Sources of Human Capital Formation

Expenditure on Education:

It is one of the most important sources of human capital formation. The education, training, and work experience of individuals determine the appropriate use of manpower. Investment in education is not only highly productive but also accelerates the economic growth of all the resources.

Expenditure on Health:

An essential component of a country's development is its health. Spending on health care directly improves a person's physical capacity and increases the pool of fit labourers. Health spending is required in the following areas:

1. Preventive medicine as curative medicines,
2. Medical intervention during the time of illness
3. The provision of clean drinking water and good sanitation is very important for improving health.

On the Job Training:

Human capital formation originates from the expenditure of on-the-job training, as it enhances labour productivity. Through practical experience and supervision from a manager, supervisor, or other experienced coworker, an employee can learn how to perform a task while on the job. Since it enables an employee to learn by carrying out his job's actual tasks, this approach may be beneficial.

Expenditure on Migration:

People move around in search of better employment opportunities. People who are unemployed in rural areas move to cities. People with technical qualifications move abroad in search of better pay. It raises living expenses and migration costs, but in comparison to the cost of migration, it increases earnings. It is therefore a source of the formation of human capital.

Expenditure on Information:

To obtain information about the labour market and other markets, expenses are incurred. It requires a significant amount of work to find information about educational institutions. Making decisions about investments in human capital and effectively utilising the stock of acquired human capital requires this information.

Reasons for Poor Human Capital Formations

Insufficient Resources

The resources allocated to the formation of human capital have been much less than the resources required to meet the educational and health needs of the country.

Serious Inefficiencies

Massive literacy, the non-education of many children, and poor health facilities are other inefficiencies that have not been adequately and properly addressed.

HUMAN CAPITAL	HUMAN DEVELOPMENT
Human capital is a narrow concept.	Human development is a broad concept.
It states education and health as means to increase labour productivity.	According to this, reading and writing skills are prerequisites for leading a healthy life, hence education and health are essential to human well-being.
It treats human beings as a means to increase productivity.	It fulfils a purpose of its own.

Growth of Education Sector In India**Pre-School Education**

Children under the age of three must attend crèches, and those between the ages of three and six must attend nurseries. Preschool facilities have been employed by the private sector in very large numbers. In order to meet the increasing demand, the government has to establish Anganwadis and Balwadis. These institutions facilitated pre-schooling for the middle class as well as the weaker sections of society.

Elementary Education

It concerns the age group of 6 to 14 years and I to VIII Standard. According to our Constitution, all children up to the age of 14 will receive free, mandatory education.

Formal Education

Formal education is classroom-based and provided by trained teachers.

Operation Blackboard

Operation Blackboard is a centrally sponsored programme that was started in 1987, immediately after the Rajiv Gandhi NPE of 1986 was released, to supply the bare minimum of crucial facilities to all primary schools in the country.

The purpose of Operation Blackboard is to ensure the provision of the minimum essential facilities in primary schools—material facilities as well as learning equipment.

Operation Blackboard envisages (a) two reasonably large rooms that are usable in all weather; (b) necessary toys and game material; (c) blackboards; (d) maps; (e) charts; and (f) other learning materials.

Non-Formal Education (6-14)

In order to achieve universal education, non-formal education was introduced for boys and girls who could not attend full-time school. Now, non-formal education is available in all states, rural, tribal, desert, hilly, remote areas, and urban slums.

Adult Education

The adult literacy rate (15+ age group) is the percentage of the population 15–24 years old who can both read and write, understanding a short, simple statement in everyday life. The adult literacy rate has shown an upward trend for females as well as males.

Secondary Education

Secondary education, which starts with classes IX and X, leads to senior secondary classes. Senior Secondary Classes XI and XII aim to develop basic skills and analytical abilities. It provides a stepping stone to higher professional and technical education.

Higher Education

The higher education system comprises both general and technical education. Higher education has undergone a manifold expansion since independence. The number of universities in the country has increased from 27 in 1950–51 to 760 in 2014–15. As of January 29, 2024, the total number of universities in India is 1,113.

The University Grants Commission (UGC) takes measures for the promotion and coordination of university education, the determination and maintenance of standards in teaching, examination, and research in universities, and the allocation and disbursement of grants to them.

Technical Education

Technical education plays a vital role in the human resources development of the country by creating skilled manpower, enhancing industrial productivity, and improving the quality of life. Since independence, there has been a phenomenal expansion of the technical education sector in the country. The All-India Council for Technical Education (AICTE) is the apex body in the field of technical education.

Indira Gandhi National Open University (IGNOU)

The Indira Gandhi National Open University (IGNOU), established by an Act of Parliament in 1985, has continuously striven to build an inclusive knowledge society through inclusive education. It has tried to increase the Gross Enrolment Ratio (GER) by offering high-quality teaching through the Open and Distance Learning (ODL) mode.

Today, it serves the educational aspirations of over 3 million students in India and other countries through 21 Schools of Studies and a network of 67 regional centres, around 2,667 learner support centres, and 29 overseas partner institutions. The university offers about 228 certificate, diploma, degree, and doctoral programmes.

Vocational Education

The greatest challenge in the Indian education system today is to provide skill-based education to the youth to reduce the mismatch in demand and supply for the skilled workforce and provide an alternative for those pursuing higher education. The Centrally Sponsored Scheme of Vocational Secondary Education at +2 level has been implemented since 1988. The scheme provides financial assistance to the states to set up administrative structures, area vocational surveys, preparation of curriculum, text books, workbook curriculum guides, training manuals, teacher training programme, strengthening of the of the technical support system for research and development, training, and evaluation, etc.

Jawahar Navodaya Vidyalaya

As per the policy of the government, one Jawahar Navodaya Vidyalaya is to be set up in each district. Accordingly, Navodaya Vidyalaya has been set up in 576 districts. Apart from that, an additional 22 Navodaya Vidyalayas were established in places where a large number of SC populations are concentrated.

1. To provide good-quality modern education to talented children from rural areas without considering their socio-economic condition
2. To ensure that all students attain a reasonable level of competence in three languages
3. To serve as focal points for improvements in the quality of school education in general through the sharing of experiences and facilities

Kendriya Vidyalaya

Founded in 1963 as Central Schools, is a network of central government schools in India that operate under the Ministry of Education. Later, the name was changed to Kendriya Vidyalaya. It is a non-profit organisation. Its schools are all affiliated to the Central Board of Secondary Education (CBSE). Its objective is to educate children of the Indian Defence Services personnel who are often posted to remote locations.

The goal of the Kendriya Vidyalayas is to guarantee that government employee children do not experience educational disadvantages as a result of their parents' relocation by offering a uniform curriculum and educational system.

The Kendriya Vidyalaya has a four - fold mission, viz.

1. To strive for excellence and establish the standard in the field of education
2. To start and encourage educational innovation and experimentation in partnership with other organisations such as the National Council of Educational Research and Training (NCERT) and the Central Board of Secondary Education (CBSE), among others.
3. To foster a feeling of oneness and the spirit of national integration in kids.
4. To meet the educational needs of children of central government personnel who are transferable, such as those in the defence and paramilitary, by offering a shared educational program.

Gender Equity

Gender equity is improving as there are less disparities in literacy rates between males and females. To raise the economic and social standing of Indian women, we must continue to support their educational opportunities. Women's education has a favourable impact on the fertility rate and health care of women and children.

Gross Enrolment Ratio (GER)

Gross Enrolment Ratio (GER) or Gross Enrolment Index (GEI) is a statistical measure used in the education sector and by the UN in its Education Index to determine the number of students enrolled in school at several different grade levels (like elementary, middle school, and high school).

$$GER = \frac{\text{number of actual students enrolled}}{\text{number of potential students enrolled}}$$

Need to Promote Women's Education

1. Women's education is essential for providing them economic independence and empowering them to save them from exploitation and domestic violence.
2. It is essential to raise the social status of women so that their quality of life can be improved.
3. Educating women helps promote family welfare programmes and, thus, population control.
4. Educated women are aware of the importance of health care and thus take proper care of their family members, contributing to human capital formation.
5. Educating a woman means educating a family. An educated woman can inculcate moral values in her children and facilitate their education.

Weakness of the Education Sector

1. **Inadequate Facilities:** Education still covers a limited population in the country. The country still has 46 percent illiterate females and 24 percent illiterate males.
2. **Unequal Accessibility:** The opportunities and benefits of education are highly unequal. For example, the education of females is far less than that of males.
3. **Irrelevant and Poor in Quality:** The quality of education in terms of its contents is low. There is too much emphasis on general education as compared to technical and vocational education.

Practice Questions

1. Classify the following into physical capital and human capital.
2. Roads, teachers, ports, engineers, technicians, scientists, factories, and buildings.
3. Consider the following cases:
4. Migration of people from rural to urban areas
5. Training is given to the employees by the firm.
6. Can you consider the above two cases as sources of human capital formation? Substantiate.
7. How does human capital formation contribute to economic growth?
8. Discuss the similarities and dissimilarities between physical capital formation and human capital formation.
9. Write an essay on “human capital formation” in India.
10. Briefly explain the sources of human capital formation.
11. Human capital is as important as physical capital for the economic development of a country. Elucidate.
12. Investment in human capital contributes to economic growth and development. Comment.
13. What are the two major sources of human capital in a country?
14. Distinguish between human capital and physical capital.
15. Trace the relationship between human capital and economic growth.
16. Define the Gross Enrolment Ratio (GER).

6. RURAL DEVELOPMENT

Rural development focusses on action for the development of areas that are lagging behind in the overall development of the village economy. Rural development covers agricultural development.

Key Issues of Rural Development

Rural development attempts to improve all aspects of rural life. Some of the areas that are challenging and need attention for development in rural India

1. Development of Human Resources
2. Land Reform
3. Development of infrastructure
4. Alleviation of Poverty
5. Development of productive resources

Development of human resources

1. **Literacy:** Especially female literacy, education, and skill development
2. **Health:** Addressing sanitation and public health

Land Reform

1. Improvement of agricultural productivity
2. Abolition of intermediaries
3. Protect the rights of the tiller of the soil.
4. Elimination and exploitation of cultivators

Development of Infrastructure

1. Development of electricity, irrigation, and transport facilities
2. Agricultural Credit, Marketing, and Financial Assistance
3. Construction of village roads and feeder roads to nearby highways
4. Facilities for agricultural research and extension

Alleviation of Poverty

1. Special measures for poverty alleviation
2. Improvement in the living conditions of a weaker section of the population
3. Develop productive employment opportunities.

Development of Productive Resources

1. Fuller utilisation of resources
2. Enhanced investment in farm areas and non-farm areas

Rural Credit

Rural credit means the provision of loans, especially in production for agriculture and non-agricultural sectors. Credit facilities in rural areas have contributed to a large increase in agricultural productivity and employment facilities in non-agricultural sectors.

Sources of Rural Credit

Rural credit availability can be broadly classified into two categories:

Non-Institution Sources: These are the traditional sources of agricultural credit in India. They include money lenders, relatives, traders, commission agents, and land lords.

Institutional Sources: They are cooperative credit, land development banks, commercial banks, regional rural banks, the government, the National Bank for Agricultural and Rural Development (NABARD), and also self-help groups.

Cooperative Credit Societies

These societies advance credit to farmers at reasonable rates of interest. These contribute nearly 30 % of the rural credit.

1. To free the farmers from the clutches of moneylenders
2. To advance credit facilities all over the country.
3. To spread credit facilities all over the country
4. To ensure the timely and continuous flow of credit to rural areas

Land Development Banks

The Land Development Bank provides long-term credit against the mortgage of land. The main objectives are:

1. To provide long-term credit
2. To provide credit for purchasing agricultural inputs, constructing permanent structures on land, etc.

Commercial Banks

After nationalisation in 1969 and 1980, commercial banks played a major role in advancing credit. The objectives of commercial banks are

1. Directly help the farmers in rural areas.
2. To provide small, medium, and long-term loans
3. To help farmers with agricultural marketing

Regional Rural Banks

Regional Rural Banks have been set up to fill the gap in credit structure in rural areas. They are assigned to open branches in those areas where there are no banking facilities available and to provide credit facilities to small and marginal framers.

National Bank For Agricultural And Rural Development (NABARD)

The Committee, formed under the chairmanship of Shri. B. Sivaraman, a former member of the Planning Commission, Government of India, in 1979, recommended the need for a new organisational device for providing undivided attention, forceful direction, and pointed focus to credit related issues linked with rural development. The National Bank for Agriculture and Rural Development (NABARD) was approved by Parliament through Act 61 of 1981, considering the recommendations.

NABARD discharge its duty by undertaking the following roles:

1. Serves as an apex financing agency for promoting the various developmental activities in rural areas.
2. Takes measures towards institution building, monitoring, formulation of rehabilitation schemes, restructuring of credit institutions, training of personnel, etc.
3. Co-ordinates the rural financing activities of all institutions engaged in developmental work at the field level
4. NABARD maintains liaison with the Government of India, state governments, the RBI, and other national-level institutions concerned with policy formulation.
5. It refinances the financial institutions that finance the rural sector.
6. It also keeps a check on its client institutions.
7. It regulates the institutions which provide financial help to the rural economy.
8. It provides training facilities to the institutions working in the field of rural uplifting
9. It regulates the cooperative banks and the RRB's.

Self-Help Groups (SGHs)

A self-help group (SHG) is a village-based financial intermediary committee usually composed of 10–20 local women or men. A mixed group is generally not preferred. Most self-help groups are located in India. Members also make small regular savings contributions over a few months until there is enough money in the group for lending. Funds may then be lent back to the members or to others in the village for any purpose. In India, many SHGs are 'linked' to banks for the delivery of Micro-Credit

Agricultural Marketing

Agricultural marketing is a process that involves the assembly, storage, processing, transportation, packaging, grading, and distribution of different agricultural commodities across the country. The independent government has adopted various measures to improve the system of agricultural marketing in the country. The following measures are adopted in order to regulate the markets:

1. **Regulated Markets:** The first measure was the regulation of markets to create an orderly and transparent marketing environment. This is organised in order to protect farmers from the malpractices of sellers and brokers.
2. **Cooperative Marketing:** Farmers form marketing societies to sell the output collectively and to take advantage of collective bargaining to obtain a better price.
3. **Infrastructural Facilities:** The government had also provided infrastructural facilities like roads, railways, warehousing, and old storage and processing units.
4. **Standardization and Grading:** Grading and quality control help farmers get a good price for the quality products they produce.
5. **Minimum Support Price:** To safeguard the Interest of the government fixes the minimum support price for agricultural products like wheat, rice, maize, cotton, sugar cane, pulses, etc. The government will willingly buy any amount of grain from the farmers at a price higher than the market price in order to help them recover their losses.

Defects of Agricultural Market in India

1. **Lack of storage facility:** Insufficient storage space for crops and food grains has resulted in product damage from rain, rodents, and insects.
2. **Distress Sale:** Most Indian farmers are poor and they have no capacity to wait for better price due to lack of storage facility
3. **Lack of Transportation:** Farmers are unable to sell their produce at a fair price because of a lack of transportation to mandis.
4. **Middleman or Intermediaries:** Increased middlemen or intermediaries in the supply chain between growers and buyers will also lower producer profits.
5. **Lack of Institutional Finance:** Insufficient institutional funding, inadequate guidance, etc. It is this that disorganises the Indian marketing system.

Measures for Improvement of Agricultural Marketing

1. Improvement in storage facilities at the farm level
2. Establishment of regulated markets.
3. Improvement of transport facilities
4. Establishment of cooperation marketing societies.
5. Provision of cheap credit
6. Provision for grading the produce to ensure good quality
7. Prompt supply of marketing information

Diversification of Agricultural Activity

This means giving gainful employment in some other allied activities in agriculture and non-farm activities to the excess of people in agriculture. This is done in order to overcome poverty, improve employment, and make rural agricultural people fully employed.

Diversification includes two aspects.

1. Diversification of crop production
2. Diversification of Productive Activities.

Non- Farm Activities or Alternative Employment Opportunity

1. Animal husbandry
2. Dairy farming,
3. Fishers
4. Horticulture
5. Poultry Farming
6. Agro-processing industries
7. Food processing industries
8. Leather industry,
9. Tourism etc.

Sustainable Development

It is the development that aims to develop the present generation without affecting the quality of life of the future generation. Sustainable development does not prohibit the use of any resources, but aims to restrict their use in such a way that they are left for future generations.

Organic Farming

Organic farming is the process of producing food naturally. This method avoids the use of synthetic chemical fertilisers and genetically modified organisms. It is very eco-friendly and very essential for sustainable development. It has zero impact on the environment.

Advantages of Organic Farming

1. It removes the use of costly agricultural inputs such as HYV seed, chemical fertilisers, pesticides, etc.
2. It generates income through exports as the demand for organically grown crops is on the rise.
3. It provides healthy food, as organically grown food has more nutritional value than food grown through chemical farming.
4. It can provide more employment opportunities in India as it requires more labourers for production than chemically produced goods.
5. Organic food is pesticide-free and is produced in an environmentally sustainable way.

Disadvantages of Organic Farming

At most, care should be taken in agricultural production. Organic farming needs to be popularised by creating awareness and willingness on the part of the farmers to adopt new methods. There is no proper infrastructure or marketing facility for these products alone. An appropriate agricultural policy should be brought in for organic farming. The yields from organic farming are lower than those from modern agricultural farming in the initial years. Therefore, small and marginal farmers may find it difficult to adapt to large-scale production.

Practice Questions

1. Analyse the role of Kudumbasree units in rural development.
2. As the majority of Indians live in rural areas, non-farm employment is more important in rural areas. Discuss the importance of non-farm employment.
3. Bring out the importance of animal husbandry, fisheries, and horticulture as sources of diversification.
4. Critically appraise rural banking operations in the light of recent farmer suicides in India.
5. Discuss the importance of credit in rural development.
6. Discuss the need for agricultural diversification in India. How far Fisheries and Horticulture can contribute to Agricultural Diversification
7. Enlist four problems faced by farmers during the initial years of organic farming.
8. Name some of the areas that need fresh initiatives for development in rural India.
9. One of the major problems faced by farmers is marketing their agricultural products. Suggest some measures taken by the government to improve agricultural marketing.
10. Organic farming has gained much importance recently. List out the benefits of organic farming.
11. Organic farming is a system of farming that maintains and restores ecological balance. Analyse this statement in light of the success of organic farming in our country.
12. Organic farming promotes sustainable development. Elucidate.
13. There is an increasing trend in Kerala's agriculture towards organic farming. List out its merits and demerits.
14. What is organic farming?
15. Why is agricultural diversification essential for sustainable development?

7. Employment: Growth, Informalisation & Other Issues

Workers and Employment

All activities that contribute to the gross national product are called economic activities. All those who are engaged in economic activities are called workers. In India, there are many

different aspects to employment. A few people work full-time during the year, while others only work a few months. Many employees do not receive adequate compensation for their labour. Simultaneously, everyone involved in economic activity is counted as employed.

Worker

A worker is a person who works for a living by engaging in productive activities. People labour to support themselves and their families by earning a living.

Casual Wage Labourer

Casual wage labour refers to employees who are not employed by their employers on a regular or long-term basis. Unlike regular employees, they are not eligible for social security benefits.

Regular Workers

Regular workers are those who have a permanent job with their employer and receive social security benefits like pensions and provident funds, among others.

Casualisation of Work Force

The process of moving from Self- Employment and regular salaried employment to casual wage work is known as casualization of workforce.

Kinds of Urban Unemployment**Industrial Unemployment**

The term "industrial unemployment" describes joblessness among those seeking employment in industrial settings.

Educated Unemployment

The term "educated unemployment" refers to the situation in which an educated individual lacks employment opportunities and is unable to find a suitable job or work for themselves.

Kinds of Rural Unemployment**Seasonal Unemployment**

Seasonal unemployment refers to job loss that happens during specific times of the year, such as in the fishing, farming, or agricultural industries.

Disguised Unemployment

When there are more people working than are actually needed, it's referred to as disguised unemployment.

Formal Sector and Informal Sector**Formal Sector**

The term "formal sector" refers to all governmental and private businesses that employ ten employees or more.

Informal Sector

The term "informal sectors" refers to all private businesses that employ fewer than ten people. For instance, labourers in agriculture, owners of small businesses, and employees on farms. Here, their income is irregular. The government reserves the right to revoke any

protection or rule at any moment. They do not keep accounts, use antiquated technology, and reside in slums.

Frictional Unemployment

Frictional unemployment is defined as unemployment that lasts for a brief period of time. Here, workers switch from one role to another.

Labour Force

Labour force = persons working + persons seeking and available for work.

(People with an age above 66 years and below 15 years, a handicapped person, and people who are not interested in working are not included.)

Work Forces

The number of people who are actually employed at a particular time is known as the work force.

Labour Force Participation Rate

The ratio of the labour force to the total population is called the labour force participation rate.

$$\text{Labour Force Participation Rate} = \frac{\text{Labour Force}}{\text{Total Population}}$$

Number of Unemployed People

$$\text{Unemployed People} = \text{Labour Force} - \text{Work Force.}$$

Worker Population Ratio

WPR refers to the participation of people in employment. It helps in knowing the proportion of the population that is actively contributing to the production of goods and services in a country.

$$\text{WPR} = \frac{\text{Total number of workers in India}}{\text{No. of work population In India}} \times 100$$

Full Employment

Full employment is a situation in which all the workers who are capable of working and willing to work get employment at a prevailing wage rate.

Employment in various Sectors of the Economy

Primary	Extraction of raw materials
	Farming and fishing
Secondary	Manufacturing
	Electricity, gas, etc.
	Construction
Tertiary	Retail
	Financial service
	Health, Education, Transportation, and Communication
	Hospitality and leisure
	Real estate
	Information technology

Growth and the Changing Structure of Employment in India: 1950–2024.

India's employment situation has changed significantly since the country gained independence in 1950. Rapid economic expansion, changes in the country's population, and technological developments have all contributed to a significant transformation in the employment market.

India's economy in the 1950s and 60s was mainly agrarian, employing more than 70% of the labour force in this sector. A gradual transition to manufacturing and services resulted from the government's emphasis on industrialisation. Economic liberalisation policies were introduced in the 1980s, which hastened this shift even more.

India's manufacturing increased from 17% to 22% of GDP between 1991 and 2011, while the services sector grew from 45% to 65% of GDP. The share of agriculture fell from 30% to 14%. A large percentage of workers were working in low-skilled, low-paying jobs as informal employment increased in tandem with this shift.

The twenty-first century has witnessed the rise of new sectors and employment opportunities in ITES, e-commerce, and IT. Millions of Indians now work as consultants, freelancers, or temporary employees, contributing to the growth of the gig economy.

However, issues like job insecurity, skill obsolescence, and regional disparities have come along with this growth. The COVID-19 pandemic has accelerated the shift towards telecommuting, automation, and digitisation, thereby requiring a workforce reskilling.

In spite of these obstacles, India's job market has advanced significantly. To encourage entrepreneurship, skill development, and digital literacy, the government has introduced programs like Made in India, Skill India, and Digital India.

To conclude, the employment structure in India has experienced a notable shift since 1950, primarily due to factors such as economic expansion, technological progress, and shifts in the country's population. Despite ongoing difficulties, the nation is well-positioned for continued growth and development, with an emphasis on inclusive employment opportunities, innovation, and skill development.

Practice Questions

1. Disguised unemployment is commonly found in the industrial sector.
2. Rajesh is an engineering graduate. But he is working as a bus conductor. Give a suitable name for such a type of unemployment.
3. Kerala has one of the highest levels of unemployment in India. At the same time, there are a large number of people working here, from other states. There is also the problem of the non-availability of workers for manual work. How will you explain this paradoxical situation?
4. Prepare a note on unemployment in India.
5. Differentiate between disguised unemployment and seasonal unemployment with the help of an example for each.
6. Define the worker population ratio.
7. Differentiate between the labour force and the work force.
8. Differentiate between the formal and informal sectors.
9. Who is a casual-wage labourer? How is it different from a regular worker?
10. Define full employment.

8. Infrastructure

Infrastructure provides supporting services in the main areas of industrial and agricultural production, domestic as well as foreign trade, and commerce. These services

include roads, railways, ports, airports, dams, power stations, oil and gas pipelines, telecommunication facilities, schools and colleges, the health system, including hospitals, the sanitary system, including clean drinking water facilities, and the monetary system, including banks, insurance, and other financial institutions.

Infrastructure can be classified into two categories: economic infrastructure and social infrastructure.

Energy, transportation, and communication are included in the economic infrastructure, whereas social infrastructure includes education, health, and housing.

Relevance of Infrastructure

The efficient functioning of a modern industrial economy depends on infrastructure. Similarly, agriculture also largely depends on it for the speedy and large-scale transport of seeds, pesticides, fertilisers, and produce using modern roadways, railways, and shipping facilities. Agriculture's production also depends on insurance and banking facilities. Infrastructure contributes to the economic development of a country both by increasing the productivity of the factors of production and by improving the quality of life of its people. Inadequate infrastructure can have multiple adverse effects on health.

The State of Infrastructure in India

Traditionally, the government has been solely responsible for developing the country's infrastructure. But it was found that the government's investment in infrastructure was inadequate. The private sector, in joint partnership with the public sector, has started playing a very important role in infrastructure development.

Some economists have projected that India will become the third-biggest economy in the world a few decades from now. For that to happen, India will have to boost its infrastructure investment. The development of infrastructure and economic development go hand in hand.

Energy

Energy is a critical aspect of the development process of a nation. It is, of course, essential for industries.

Sources Of Energy:

Conventional Sources of Energy:

There are commercial and non-commercial sources of energy.

Commercial sources are coal, petroleum, and electricity, as they are bought and sold. Commercial sources of energy are generally exhaustible.

Non-commercial sources of energy are firewood, agricultural waste, and dried dung. These are non-commercial, as they are found in nature. Non-commercial sources are generally renewable.

Both commercial and non-commercial sources of energy are known as conventional sources of energy.

Non-Conventional Sources of Energy

There are three other sources of energy: solar energy, wind energy, and tidal power.

Power/Electricity

The most visible form of energy, which is often identified with progress in modern civilisation, is power, commonly called electricity. It is a critical component of infrastructure that determines the economic development of a country.

In India, in 2016, thermal sources accounted for 67 percent of the power generation capacity. Hydel power accounted for 14 percent, while nuclear power accounted for only 2 percent. India's energy policy encourages two energy sources: - Hydel and wind.

Some Challenges in the Power Sector

Some of the challenges that India's power sector faces today are:

1. India's installed capacity to generate electricity is not sufficient to feed an annual economic growth of 7–8 percent.
2. State Electricity Boards (SEBs), which distribute electricity, incur losses. This is due to transmission and distribution losses, wrong pricing of electricity, and other inefficiencies.
3. Private sector power generators are yet to play their role in a major way.
4. There is general public unrest due to high power tariffs and prolonged power cuts in different parts of the country.
5. Thermal power plants, which are the mainstay of India's power sector, are facing shortages of raw materials and coal supplies.

Health and the State of Health Infrastructure

1. Health is a vital public good and a basic human right.
2. Health is not only the absence of disease but also the ability to realise one's potential.
3. Health is a holistic process related to the overall growth and development of the nation.
4. The development of health infrastructure ensures a country has healthy manpower for the production of goods and services.
5. Health infrastructure includes hospitals, doctors, nurses, and other paramedical professionals; beds; equipment required in hospitals; and a well-developed pharmaceutical industry.
6. The government has the constitutional obligation to guide and regulate all health-related issues, such as medical education, adulteration of food, drugs, and poisons, the medical profession, vital statistics, mental deficiency, and lunacy.
7. The Union Government evolves broad policies and plans through the Central Council of Health and Family Welfare.
8. At the village level, a variety of hospitals, technically known as primary health centres (PHCs), have been set up by the government.
9. In India, a large number of hospitals are run by voluntary agencies and the private sector.
10. In recent times, the private sector has been playing a dominant role in medical education and training, medical technology and diagnostics, the manufacture and sale of pharmaceuticals, etc.
11. Owing to the liberalisation measures, many non-resident Indians and industrial and pharmaceutical companies have set up super-speciality hospitals to attract India's rich and medical tourists.

Indian Systems of Medicine (ISM):

It includes six systems: Ayurveda, Yoga, Unani, Siddha, Naturopathy, and Homoeopathy (). ISMs have huge potential and can solve a large part of our healthcare problems because they are effective, safe, and inexpensive.

Indicators of Health and Health AYUSH Infrastructure—A Critical Appraisal:

1. The health status of a country can be accessed through indicators such as infant mortality and maternal mortality rates, life expectancy, and nutrition levels, along with the incidence of communicable and non-communicable diseases.
2. One study points out that India has about 17 percent of the world's population, but it bears a frightening 20 percent of the global burden of diseases (GBD).

***GBD** is an indicator used by experts to gauge the number of people dying prematurely due to a particular disease, as well as, the number of years spent by them in a state of 'disability' owing to the disease.*

3. In India, more than half of GBD is caused by communicable diseases such as diarrhoea, malaria, and tuberculosis.
4. Every year, around five lakh children die of water-borne diseases. The danger of AIDS is also looming large.
5. Though 70 percent of India's population lives in rural areas, only one-fifth of its hospitals are located in rural areas. Rural India has only about half the number of dispensaries.
6. The Public Health Centres located in rural areas do not even offer X-ray or blood testing facilities.
7. Villagers have no access to any specialised medical care, like paediatrics, gynaecology, anaesthesia, and obstetrics.
8. The shortage of doctors in rural areas still persists. While one-fifth of these doctor graduates leave the country for better monetary prospects, many others opt for private hospitals, which are mostly located in urban areas.

Women's Health

Women constitute about half of the total population in India. They suffer many disadvantages as compared to men in the areas of education, participation in economic activities and healthcare. More than 50 percent of married women in the age group of 15–49 years have anaemia and nutritional anaemia caused by iron deficiency, which has contributed to 19 percent of maternal deaths.

The deterioration in the child sex ratio in the country from 927 in 2001 to 914 in 2011 points to the growing incidence of female foeticide. Abortions are also a major cause of maternal morbidity and mortality in India.

Schemes of Government of India to improve women's health**Surakshit Matritva Aashwasan (SUMAN)**

The scheme aims to provide dignified and quality health care at no cost to every woman and newborn visiting a public health facility in order to bring down maternal and infant mortality rates in the country. The government will also provide free transport from home to health institutions.

Janani Suraksha Yojana (JSY)

It is a demand promotion and conditional cash transfer scheme. It was launched in 2005 with the objective of reducing maternal and infant mortality by promoting institutional delivery among pregnant women.

Janani Shishu Suraksha Karyakram (JSSK)

It aims to eliminate out-of-pocket expenses for pregnant women and sick infants by entitling them to free delivery, including caesarean, free transport, diagnostics, medicines, other consumables, diet, and blood in public health institutions.

Pradhan Mantri Surakshit Matritva Abhiyan (PMSMA)

It was launched in 2016. It provides pregnant women with free and quality-assured pregnancy care on the 9th day of every month. The program ensures that women in their second or third trimesters of pregnancy receive a minimum package of pregnancy care services at designated government health facilities.

Pradhan Mantri Matru Vandana Yojana (PMMVY)

It came into effect in 2017. It is a direct benefit transfer (DBT) scheme under which cash benefits are provided to pregnant women in their bank accounts directly to meet enhanced nutritional needs and partially compensate for wage loss.

Labour Room Quality Improvement Initiative (LaQshya):

It was launched in 2017. It aims to improve the quality of care in labour rooms and maternity operation theatres to ensure that pregnant women receive respectful and quality care during delivery and the immediate postpartum period.

Practice Questions

1. Classify the following into economic infrastructure and social infrastructure
Roads, Railways, Housing, Capitals, Power, Schools, Colleges, and Ports
2. Infrastructure is an important base for the development of the nation. Do you agree? Substantiate.
3. Give examples for “social infrastructure.”
4. Compare and contrast the terms economic and social infrastructure. Is there any inequality in rural and urban infrastructure?
5. Infrastructure needs to be upgraded from time to time to maintain high economic growth. Do you agree with this statement? Justify your answer.
6. Mention two government organisations each that regulate the health and education sectors.
7. Our health infrastructure facilities are insufficient to take care of our total population. Comment?
8. Distinguishes between Primary health care and secondary health care.
9. “Any investment in education and health is unproductive if it does not enhance the output of goods and services.” Write your opinion about this statement.
10. What do you mean by AYUSH?

9. . ENVIRONMENT AND SUSTAINABLE DEVELOPMENT

Environment — Definition and Functions

Environment is defined as the total planetary inheritance and the totality of all resources. It includes all the biotic and abiotic factors that influence each other. *Living elements*: the birds, animals and plants, forests, fisheries etc. are biotic elements; *abiotic elements* include air, water, land etc. Rocks and sunlight are all examples of abiotic elements of the environment.

Functions of the Environment

The environment performs four vital functions:

Environment supplies resources

(Resources include both renewable and non-renewable resources. Renewable resources are those which can be used without the possibility of the resource becoming depleted or exhausted. That is, a continuous supply of the resource remains available. Examples of renewable resources are the trees in the forests and the fishes in the ocean. Non-renewable resources, on the other hand, are those which get exhausted with extraction and use, for example, fossil fuel.)

Environment assimilates waste

The environment possesses an incredible ability to assimilate waste. This concept is known as assimilative capacity. When we add biodegradable waste to the soil, such as fallen leaves or organic waste, the stream's bacteria get to work. To help break down this waste, they use oxygen from the atmosphere.

Environment sustains life by providing genetic and bio diversity

These ecosystems give us vital services that keep us alive. They provide food, maintain a stable climate, and clean our water and air. Because of natural processes and human interactions with the environment, biodiversity has developed over billions of years of evolution.

Environment also provides aesthetic services like scenery etc.

Environmental gifts include aesthetic services. Our quality of life is improved by these picturesque views, whether it be of a calm beach, a rugged mountain range, or a peaceful forest.

The environment generally had two capacities.: Carrying Capacity And Absorptive Capacity

Carrying Capacity

It means resource extraction is not above the rate of regeneration of the resource, and the waste generated is within the assimilating capacity of the environment.

Absorptive Capacity

It means the ability of the environment to absorb degradation.

Global Warming

Global warming is a gradual increase in the average temperature of the earth's lower atmosphere as a result of the increase in greenhouse gases since the Industrial Revolution. Global warming is human-induced. It is caused by man-made increases in carbon dioxide and other greenhouse gases through the burning of fossil fuels and deforestation. Some of the longer-term results of global warming are the melting of polar ice with a resulting rise in sea level and coastal flooding, the extinction of species, frequent tropical storms, and an increased incidence of tropical diseases. A UN Conference on Climate Change, held in Kyoto, Japan, in 1997, resulted in an international agreement to fight global warming.

Ozone Depletion

Ozone depletion refers to the phenomenon of reductions in the amount of ozone in the stratosphere. The problem of ozone depletion is caused by high levels of chlorine and bromine compounds in the stratosphere. The origins of these compounds are chlorofluorocarbons (CFC), used as cooling substances in air conditioners and refrigerators. As a result of the depletion of the ozone layer, more ultraviolet (UV) radiation comes to Earth and causes damage to living organisms. UV radiation seems responsible for skin cancer in human beings.

State of India's Environment

1. India has abundant natural resources in terms of rich quality soil, hundreds of rivers and tributaries, lush green forests, plenty of mineral deposits beneath the land surface, a vast stretch of the Indian Ocean, ranges of mountains, etc.
2. The Indo-Gangetic plains, spread from the Arabian Sea to the Bay of Bengal, are one of the most fertile, intensively cultivated, and densely populated regions in the world.
3. Large deposits of iron ore, coal, and natural gas are found in the country.
4. Air pollution, water contamination, soil erosion, deforestation, and wildlife extinction are some of the most pressing environmental concerns in India.
5. The priority issues identified are (i) land degradation, (ii) biodiversity loss, (iii) air pollution, with special reference to vehicular pollution in urban cities, (iv) the management of fresh water, and (v) solid waste management.

Factors Responsible for Land Degradation

1. Loss of vegetation occurring due to deforestation
2. Unsustainable fuel wood and fodder extraction
3. Shifting cultivation
4. Encroachment into forest lands
5. Forest fires and overgrazing
6. Non-adoption of adequate soil conservation measures
7. Improper crop rotation
8. Indiscriminate use of Agro-chemicals such as fertilisers and pesticides
9. Improper planning and management of irrigation systems
10. Extraction of groundwater from competing uses of land for forestry, agriculture, pastures, human settlements, industries, etc.

Sustainable Development

Sustainable development aims at promoting the kind of development that minimises environmental problems and meets the needs of the present generation without compromising the ability of the future generation to meet their own needs.

1. The concept of sustainable development was emphasised by the United Nations Conference on Environment and Development (UNCED), which defined it as 'development that meets the needs of the present generation without compromising the ability of the future generation to meet their own needs'.
2. Sustainable development aims at decreasing the absolute poverty of the poor by providing lasting and secure livelihoods that minimise resource depletion, environmental degradation, cultural disruption, and social instability.
3. Sustainable development is a development that meets the basic needs of all, particularly the poor majority, for employment, food, energy, water, and housing, and ensures the growth of agriculture, manufacturing, power, and services to meet these needs.

The Brundtland Commission places emphasis on protecting the future generation. The present generation can promote development that places emphasis on (i) the conservation of natural assets, (ii) the preservation of the regenerative capacity of the world's natural ecological system, and (iii) avoiding the imposition of added costs or risks on future generations.

According to Herman Daly, a leading environmental economist, to achieve sustainable development, the following needs to be done:

1. Limiting the human population to a level within the carrying capacity of the environment
2. Renewable resources should be extracted on a sustainable basis; that is, the rate of extraction should not exceed the rate of regeneration.

3. For non-renewable resources, the rate of depletion should not exceed the rate of creation of renewable substitutes.
4. Inefficiencies arising from pollution should be corrected.

Strategies for Sustainable Development

Use of Non-conventional Sources of Energy

India is hugely dependent on thermal and hydropower plants to meet its power needs. Both of these have adverse environmental impacts. Wind power and solar rays are good examples of non-conventional energy sources, and they are more eco-friendly.

LPG, Gobar Gas in Rural Areas

Households in rural areas generally use wood, dung cake, or other biomass as fuel. This practice has several adverse implications, like deforestation, a reduction in green cover, the wastage of cattle dung, and air pollution. To rectify the situation, LPG at the Gobar gas plant is being provided through easy loans and subsidies. It reduces household pollution to a large extent. Also, energy waste is minimised.

CNG in Urban Areas

In India, the use of compressed natural gas (CNG) as fuel in public transport systems has significantly lowered air pollution.

Wind Power

In areas where the speed of wind is usually high, wind mills can provide electricity without any adverse impact on the environment.

Solar Power through Photovoltaic Cell

India is naturally endowed with a large quantity of solar energy in the form of sunlight. Now, with the help of photovoltaic cells, solar energy can be converted into electricity. These cells use a special kind of material to capture solar energy and then convert the energy into electricity. This technique is also totally free from pollution.

Mini-Hydel Plants

In mountainous regions, streams can be found almost everywhere. By using these streams, we can construct mini-hydel power plants. Such power plants are environment-friendly as they do not change the land use pattern in areas where they are located, and they generate enough power to meet local demands.

Traditional Knowledge and Practices

Traditionally, Indian people have been close to their environment. They practiced an environment-friendly agriculture system, healthcare system, housing, transport, etc.

Bio composting

With the increase in agricultural production, we completely switched over to chemical fertilisers. It adversely affected our land, health, and water bodies due to chemical contamination. To regain the quality of land, farmers have started using compost made from organic wastes of different types. Earthworms can convert organic matter into compost faster than the normal composting process.

Bio pest Control

With the advent of the green revolution, our farmers are using increasingly chemical pesticides for higher yields. As a result, food products, soil, water bodies, and even groundwater were polluted with pesticides. Even milk, meat, and fish were also contaminated. To meet these challenges, better methods of eco-friendly pest control measures are now being used. One such

step is the use of pesticides based on plant products. Mixed cropping and growing different crops in consecutive years on the same land have also helped farmers.

Practice Questions

1. What are the strategies for sustainable development?
2. Information technology plays a vital role in achieving sustainable development and food security. Comment.
3. Do you think that the present pattern of development is sustainable? List out the major issues that make development unsustainable
4. What are the functions of the environment?
5. The important issues related to sustainable development are resource depletion and environmental pollution. Give two reasons each for this problem that you observe in your locality. Suggest remedial measures.
6. Prepare an essay on the importance of organic farming in promoting sustainable development.
7. Development that meets the needs of the present generation without compromising the ability of the future generation to meet their own needs is sustainable development. State the issues related to sustainable development in light of the above statement.
8. Land degradation is a major threat to sustainable development. Prepare a note on the recent landslide that happened in Kerala.
9. Give your suggestions for achieving sustainable development in our country.
10. The Copenhagen Climate Change Conference in December 2009 expressed great anxiety over global warming in the world. How does this phenomenon affect India and its sustainable development?

10. DEVELOPMENT EXPERIENCES OF INDIA & ITS NEIGHBOURS

Nations have been primarily trying to adopt various means which will strengthen their own domestic economies. To this effect, they are forming regional and global economic groupings such as the SAARC, European Union, ASEAN, G-8, G-20, BRICS etc.

Developmental Path: India, China, and Pakistan

All three nations have started on their developmental paths at the same time.

1. India and Pakistan became independent nations in 1947 and The People's Republic of China was established in 1949.
2. The three countries had started their planning strategies in similar ways.
3. India announced its first five-year plan for 1951–56, Pakistan announced its first five-year plan, now called the Medium-Term Development Plan, in 1956. China announced its first five-year plan in 1953.
4. India and Pakistan adopted similar strategies, such as creating a large public sector and raising public expenditure on social development.
5. All three countries had similar growth rates and per capita incomes until the 1980s.

Development Strategies of China

Great Leap Forward

China began its Great Leap Forward campaign in 1958 with the goal of industrialising the nation extensively. It was encouraged for people to start large-scale businesses in their own backyards.

Great Proletarian Cultural Revolution (1966-1976)

Mao Se Tung began a broad cultural revolution in 1965. Professionals and students were sent to the countryside to work and study during the revolution.

1978 Reforms

China started gradually enacting numerous reforms in 1978. The agricultural, international trade, and investment sectors have all adopted these reforms. China's economic reforms aimed to produce enough surplus.

One-Child Norm in China

The one-child norm in China was introduced in 1970 to reduce the growth of the population. China has successfully reduced the growth rate of its population. But after a few decades in China, there were more elderly people in proportion to young people. This forced China to take steps to provide social security measures (pension scheme, old age home, etc.) with fewer workers.

Development Strategies of Pakistan

Mixed Economy

Pakistan has embraced a mixed economy. That is, the public and private sectors exist side by side.

Import Substitution

Pakistan established a regulatory framework for import substitution in the late 1950s and early 1960s. The policy combined direct import controls on rival imports with tariff protection for companies that manufacture consumer goods.

Green Revolution:

Pakistan adopted the green revolution to boost food grain output and boost food productivity and self-sufficiency. This had a major effect on Pakistan's economy's agrarian system.

Demographic Profile of India, China, and Pakistan: A Comparison

Birth rate /1000

Country	2017	2024 est.
India	18.08	16.2
China	12.43	10.2
Pakistan	28.60	25.5

Death Rate /1000

Country	2017	2024 est.
India	7.21	9.1
China	7.11	7.7
Pakistan	6.99	5.9

Density of Population/ KM²

Country	2017	2023
India	450	435
China	148	149
Pakistan	270	302

Growth Rate of Population

Country	2017	2024 est.
India	1.06%	0.72%
China	0.56%	0.23%
Pakistan	2.07%	1.86%

Sex Ratio Male/ 1000 Female

Country	2017	2024 est.
India	1100	1060
China	1130	1040
Pakistan	1090	1050

Infant Mortality Rate/ 1000 Live Births

Country	2017	2024 est.
India	31.50	30.4
China	7.90	6.2
Pakistan	58.80	51.5

**Maternal Mortality Rate
(deaths /100,000 live births)**

Country	2017	2020.
India	145	103
China	29	23
Pakistan	140	154

HDI Ranking

Country	2017	2024 est.
India	130	134
China	86	75
Pakistan	150	164

Below Poverty Line

Country	2017
India	23%
China	24.3%
Pakistan	3.1%

Literacy Rate

Country	2017	2024 est.
India	74.04%.	74.4%.
China	96.8 %	98.5%.
Pakistan	58%	58%.

Common success story of India & Pakistan

6. A substantial rise in GDP per capita
7. Self-sufficiency in food production
8. The dualistic nature of the economy is gradually declining.
9. There is a considerable decline in the incidence of poverty.

Common failures of India and Pakistan

10. Relatively slow pace of GDP growth rate as compared to China
11. Poor performance in the HDI ranking
12. Poor fiscal management
13. Lack of political interest in social prosperity

Areas where India has an edge over Pakistan

1. Skilled manpower
2. Investment in Education
3. Health facilities

Areas where Pakistan has an edge over India

4. Increased Urbanisation
5. A low percentage of the population is below the poverty line.
6. A higher percentage of the population has access to improved sanitation facilities.

Areas where China has an edge over India

7. China has successfully focused on Pro-poor reforms.
8. Agrarian reforms have been effectively carried out.
9. Export-driven manufacturing has significantly grown, adding to the pace of GDP growth.
10. SEZ policy has proved to be a boon for FDI flow in China.

Practice Questions

1. The policy measures adopted in India and China are almost similar. But the impacts of these measures were different in these two economies. Compare the development experiences of these economies.
2. China's rapid economic growth can be traced back to its reforms in 1978. Do you agree? Justify your answer.
3. Compare the demographic features of India, China, and Pakistan.
4. Evaluate the various factors that led to the rapid growth of economic development in China.
5. Do you think it is necessary for India and Pakistan to concentrate on the manufacturing sector as China does? Why?
6. Pakistan has successfully reduced the percentage of people below the poverty line, but
7. The percentage of people below the poverty line in India is still high. State any two reasons for it.
8. Group the following features pertaining to the economies of India, China, and Pakistan under three heads: (One-child norm, low fertility rate, high degree of urbanisation, mixed economy, very high fertility rate, large population, high density of population, growth due to the manufacturing sector, growth due to the service sector)
9. Discuss the differences in the political systems of India, China, and Pakistan.
10. Write a short note on the development strategies of China: Great Leap Forward
11. What is the important implication of the 'one child norm' in China?
12. Do you think import substitution can protect domestic industries? Give your views in relation to India, China, and Pakistan's development experiences.

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